

Integrating Genomic Data: More Annotation Tools and Interpretation

HUGEN 2073

Genomic Data Visualization and Integration

Jon Chernus

Last time

- Querying LD information from 1000 Genomes Project with LDlink
 - Using the R package LDlinkR to retrieve LD information
 - Using LDlinkR to combine LD information with other annotation
 - e.g., find SNPs in LD with a given set of SNPs
 - e.g., find eQTLs, prior GWAS associations in LD with a given set of SNPs
 - Flexibility in specifying population, strength of LD, and distance
- VEP (Variant Effect Predictor)
 - Online (or command line) tool for mapping variants to transcripts and other genome elements
 - Genic/exonic/intronic/regulatory consequences
 - Predicted deleteriousness (PolyPhen, SIFT, CADD)
 - More annotations
- Querying variants for regulatory function
 - HaploReg: query variants in LD for enhancer/eQTL/evidence of regulatory role
 - RegulomeDB: similar to HaploReg
 - Quantifies “probability” of a variant having regulatory function
 - “Ranks” variants according to qualitative evidence strength
 - HaploReg/RegulomeDB can be accessed with the R package HaploR (which is a little bug-gy)

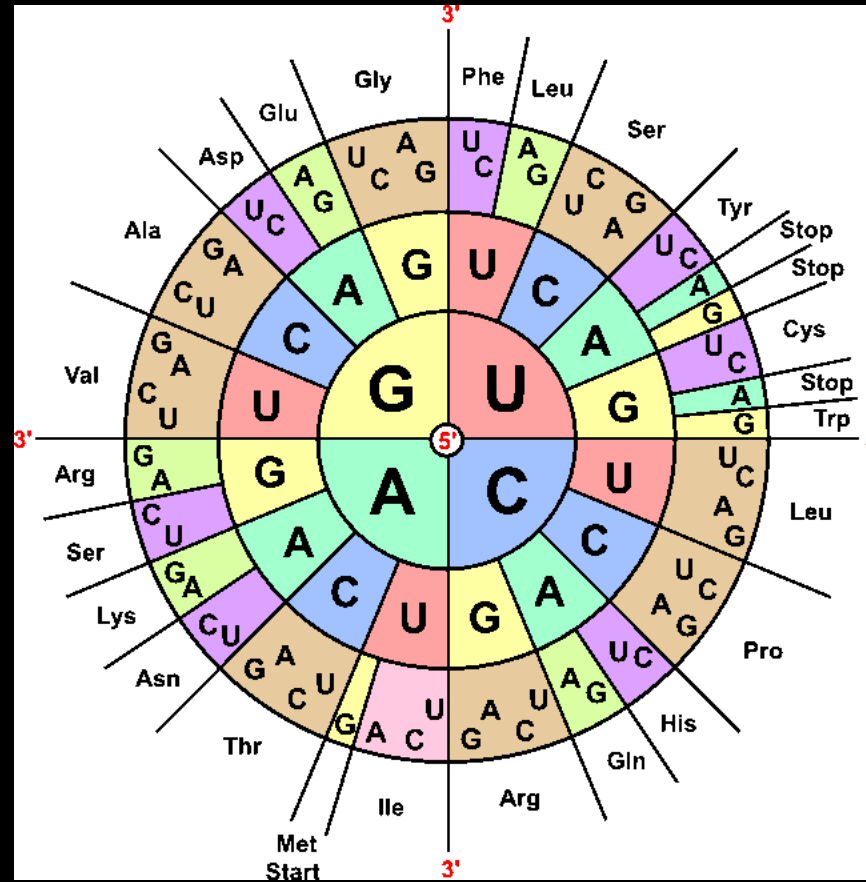
This lecture

- More detailed discussion of variant “consequences”
 - Review of coding/genic variant terminology
 - Review of amino acid substitution
 - Meaning of SIFT/PolyPhen/CADD scores
 - “Clinical” phenotype associations
 - OMIM
 - ClinVar (/ClinGen)
 - Using VEP, variation browser, and UCSC Genome Browser to query these
- “High-throughput” annotation with ANNOVAR
- “Interpretation” of annotations/evidence

Classifying genic variation

- Exonic
 - Synonymous
 - Missense
 - Stop-gain/nonsense
 - Stop-loss
 - Indel/frameshift
 - (Synonymous exonic variants can alter splicing, too – preventing “normal” splicing or adding “abnormal” splicing)
 - UTR variation
- Intronic
 - Splice site

Codons



Synonymous variants

Don't change amino acid

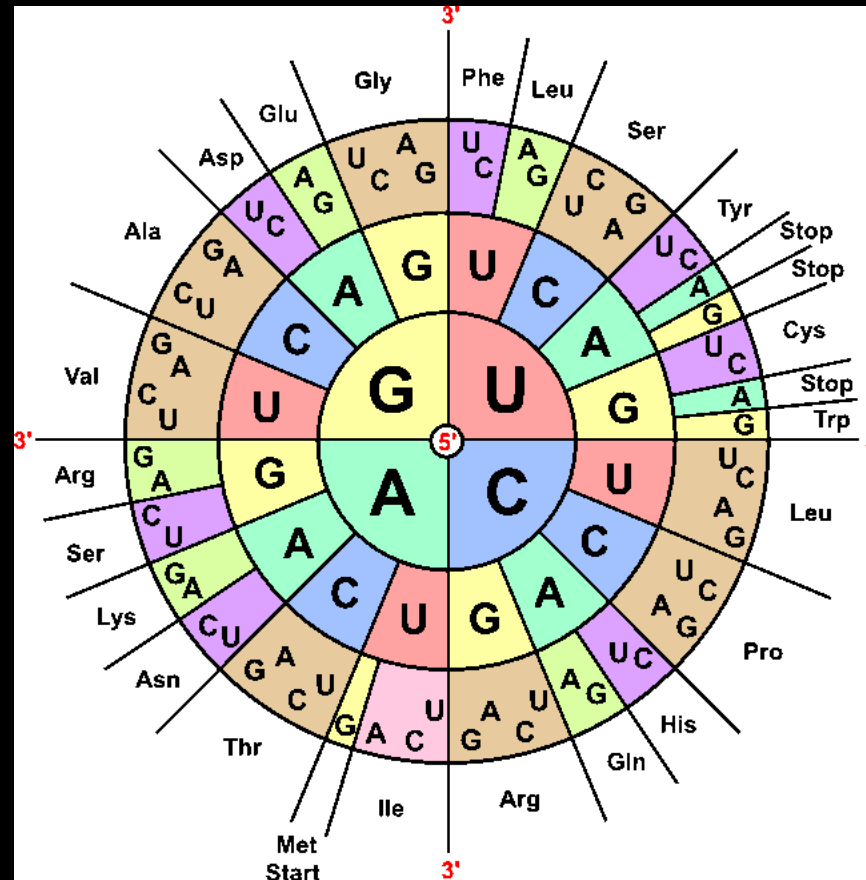
Examples:

GAG to GAA

AGC to AGU

UUA to CUA

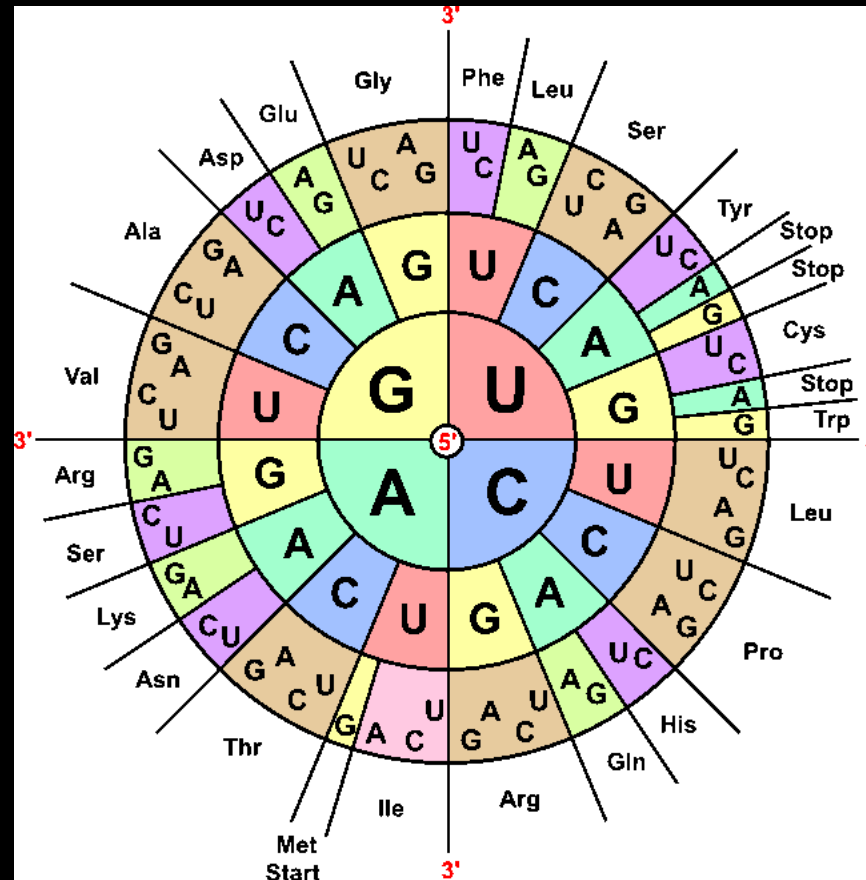
Synonymous SNPs may still matter (e.g., by affecting protein binding or splicing)



Missense variants

Substitute one amino acid
for another

Examples:
GAG to GAC
AGA to ACA
CCA to UCA



Amino acids

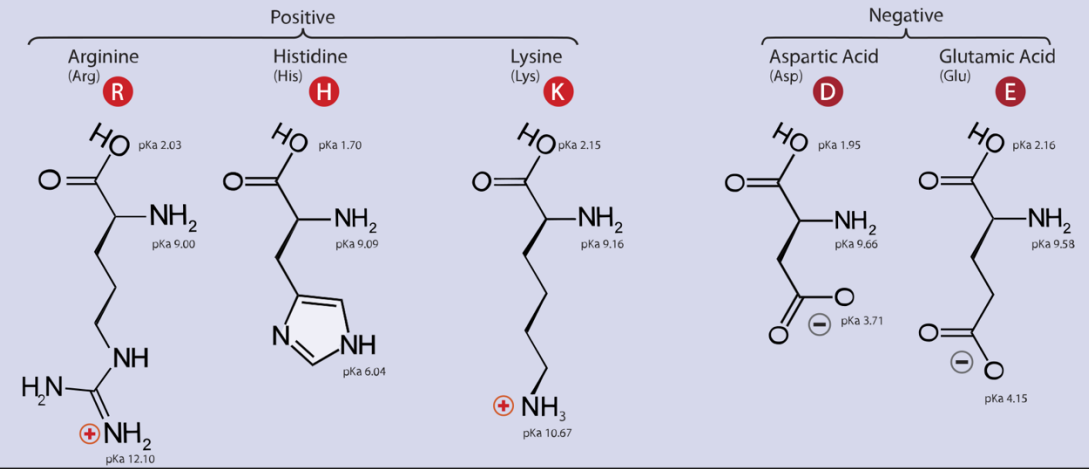
Not all amino acid substitutions are equally significant.

Some may be well-tolerated, while others may be highly deleterious.

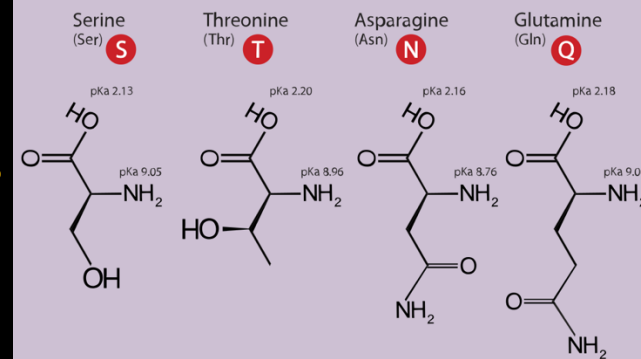
Twenty-One Amino Acids

⊕ Positive
⊖ Negative
• Side chain charge at physiological pH 7.4

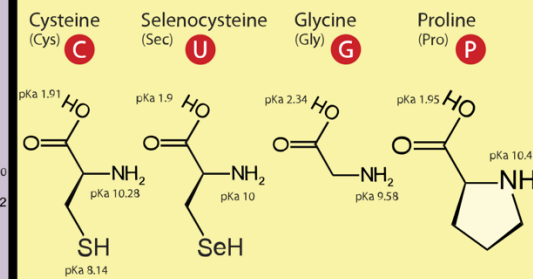
A. Amino Acids with Electrically Charged Side Chains



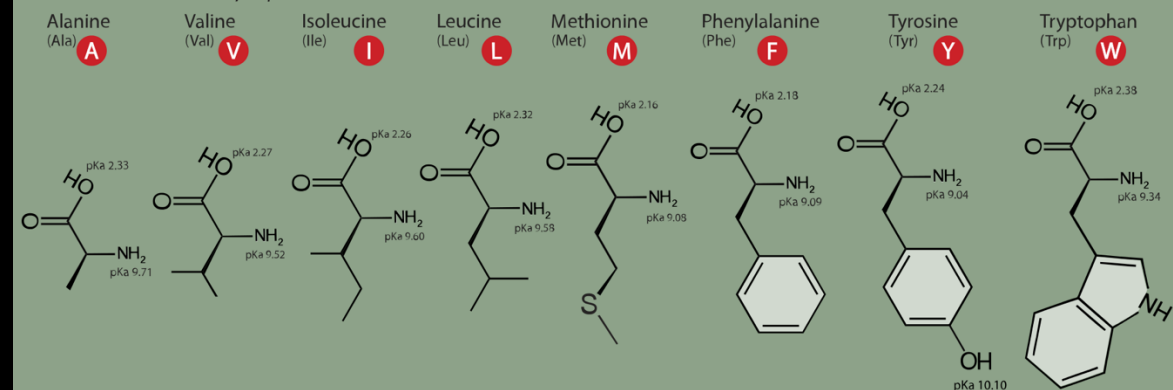
B. Amino Acids with Polar Uncharged Side Chains



C. Special Cases



D. Amino Acids with Hydrophobic Side Chain



Missense variation

SIFT score (Sorting Intolerant From Tolerant)
Predicts deleteriousness of amino acid substitutions
Uses position and conservation (not structure)

SIFT has been applied to human variant databases and was able to distinguish mutations involved in disease from neutral polymorphisms (3). Assuming that disease-causing amino acid substitutions are damaging to protein function, we applied SIFT to a database of missense substitutions associated with or involved in disease (4). SIFT predicted 69% to be damaging. When SIFT was applied to the non-synonymous SNPs in dbSNP (5), a database of putative SNPs, 25% of the variants were predicted to be deleterious. This was similar to SIFT's 20% false positive error which suggested that most non-synonymous SNPs are functionally neutral. Furthermore, a subset of the variants from dbSNP predicted to affect function were involved in disease which confirmed SIFT sensitivity.

The SIFT algorithm relies solely on sequence for prediction, yet performs similarly to tools that use structure (3,6–8). An advantage of not requiring structure is that a larger number of substitutions can be predicted on. Of the non-synonymous SNPs identified by the SNP Consortium, 74% were sufficiently similar to homologs in protein sequence databases for SIFT prediction. The number of substitutions that SIFT can predict on is expected to increase as more genomes are sequenced and more protein sequences become available.

Output: number between 0 and 1 (smaller=predicted “more intolerant”/more deleterious, larger=more likely benign)
Score < 0.05 is a common cutoff for predicting probably “damaging”/“deleterious”
Recall we can access SIFT scores through VEP

SIFT: predicting amino acid changes that affect protein function

Pauline C. Ng and Steven Henikoff^a

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<https://sift.bii.a-star.edu.sg>

The screenshot shows the SIFT web interface with a table of variants and a CADD score overlay. The table has columns for Variant ID, Chr: bp, Alleles, Evidence, AA, AA coord, and SIFT. The CADD overlay shows a color scale from 0 (green) to 1 (red) with a slider and an 'Apply' button.

Variant ID	Chr: bp	Alleles	Evidence	AA	AA coord	SIFT
rs1437042753	3:25609282	G/A	gnomAD	R/W	1327	0
rs1246519283	3:25624786	G/A	gnomAD	R/W	743	0
rs1173166728	3:25627237	G/A	gnomAD	R/C	651	0
rs1357930313	3:25632778	T/C	gnomAD	D/G	343	0
rs776876327	3:25634002	G/A/T	gnomAD	R/C	284	0.01
rs376036396	3:25615236	T/A	gnomAD	E/V	1182	0.01
rs1349960311	3:25615269	A/G	gnomAD	L/P	1171	0

Missense variation

PolyPhen

<http://genetics.bwh.harvard.edu/pph2/>

Prediction

PolyPhen-2 predicts the functional significance of an allele replacement from its individual features by Naïve Bayes classifier trained using supervised machine-learning.

Two pairs of datasets were used to train and test PolyPhen-2 prediction models. The first pair, **HumDiv**, was compiled from all damaging alleles with known effects on the molecular function causing human Mendelian diseases, present in the UniProtKB database, together with differences between human proteins and their closely related mammalian homologs, assumed to be non-damaging. The second pair, **HumVar**, consisted of all human disease-causing mutations from UniProtKB, together with common human nsSNPs (MAF>1%) without annotated involvement in disease, which were treated as non-damaging.

The user can choose between HumDiv- and HumVar-trained PolyPhen-2 models. Diagnostics of Mendelian diseases requires distinguishing mutations with drastic effects from all the remaining human variation, including abundant mildly deleterious alleles. Thus, HumVar-trained model should be used for this task. In contrast, HumDiv-trained model should be used for evaluating rare alleles at loci potentially involved in complex phenotypes, dense mapping of regions identified by genome-wide association studies, and analysis of natural selection from sequence data, where even mildly deleterious alleles must be treated as damaging.

For a mutation, PolyPhen-2 calculates Naïve Bayes posterior probability that this mutation is damaging and reports estimates of false positive rate (FPR, the chance that the mutation is classified as damaging when it is in fact non-damaging) and true positive rate (TPR, the chance that the mutation is classified as damaging when it is indeed damaging). A mutation is also appraised qualitatively, as **benign**, **possibly damaging**, or **probably damaging** based on pairs of false positive rate (FPR) thresholds, optimized separately for each model (e.g., HumDiv and HumVar).

Current version 2.2 of the PolyPhen-2 uses 5% / 10% FPR for **HumDiv** model and 10% / 20% FPR for **HumVar** model as the thresholds for this ternary classification. Mutations with their posterior probability scores associated with estimated false positive rates at or below the first (lower) FPR value are predicted to be **probably damaging** (more confident prediction). Mutations with the posterior probabilities associated with false positive rates at or below the second (higher) FPR value are predicted to be **possibly damaging** (less confident prediction). Mutations with estimated false positive rates above the second (higher) FPR value are classified as **benign**.

If the lack of data does not allow to make a prediction then the outcome is reported as **unknown**.

PolyPhen-2 is an automatic tool for prediction of possible impact of an amino acid substitution on the structure and function of a human protein. This prediction is based on a number of features comprising the sequence, phylogenetic and structural information characterizing the substitution.

For a given amino acid substitution in a protein, PolyPhen-2 extracts various sequence and structure-based features of the substitution site and feeds them to a probabilistic classifier.

PolyPhen

Scores range from 0 to 1.

Closer to 0 = predicted more likely to be benign

Close to 1 = predicted more likely to be deleterious

Access through web tool or VEP

Query Data

Protein or SNP identifier

SLC39A10

Protein sequence
in FASTA format

Position

2

Substitution

AA₁ A R N D C E Q G H I L **K** M F P S T W Y V
AA₂ A R N D C E Q G H I L K M F P S T W Y V

Query description

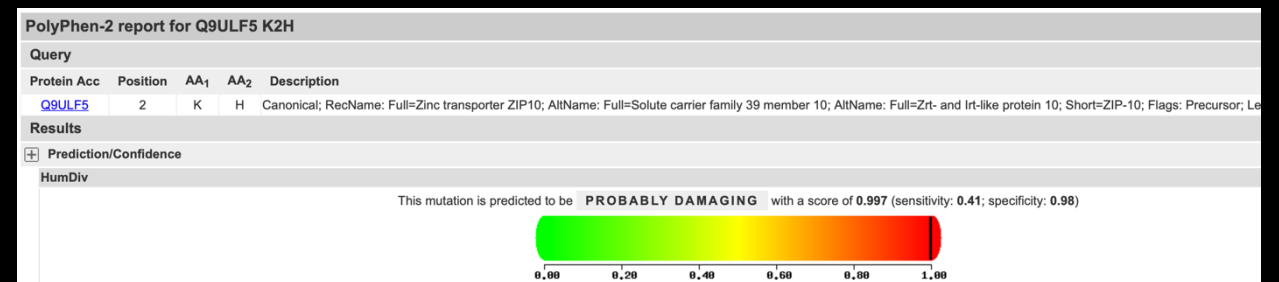
Click to select His
(Histidine)

Submit Query

Clear

Check Status

Display advanced query options



Stop-gain/nonsense variants

Nonsense mutations refer to gain of a premature stop codon, resulting in a truncated protein (or may not be translated).

Examples:

GAG to UAG

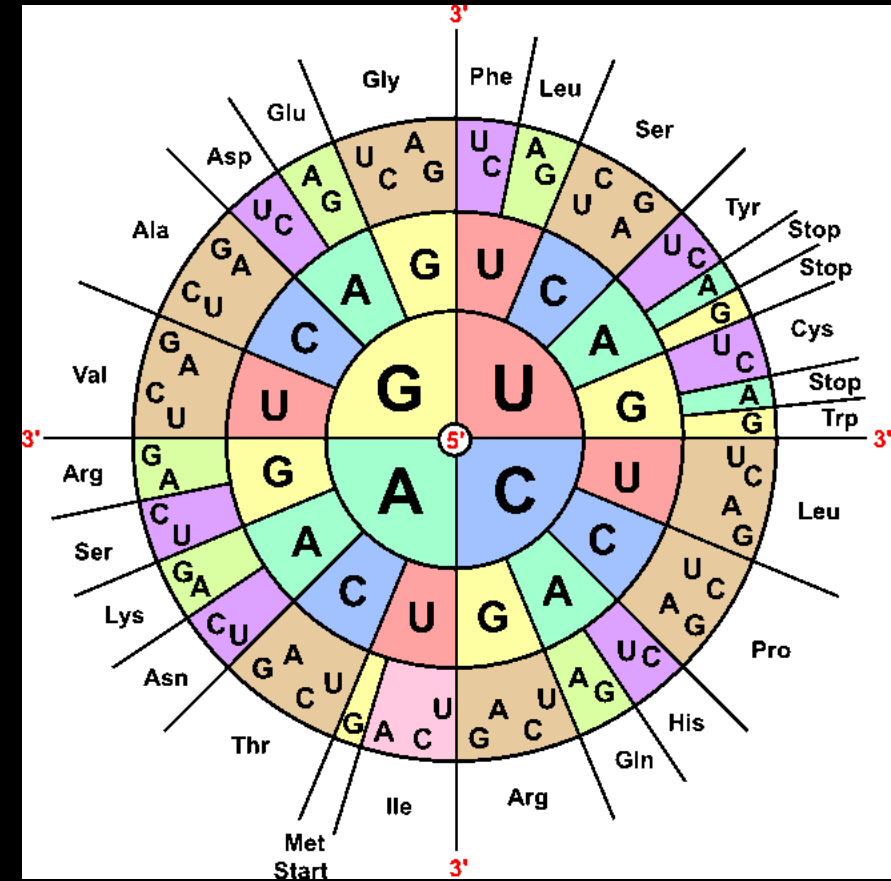
UCA to UAA

UGG to UGA

Losing a start codon is also possible.

Examples:

AUG to CUG



Stop-loss and start-gain variants

Stop-loss examples:

UGA to UGC

UAA to UCA

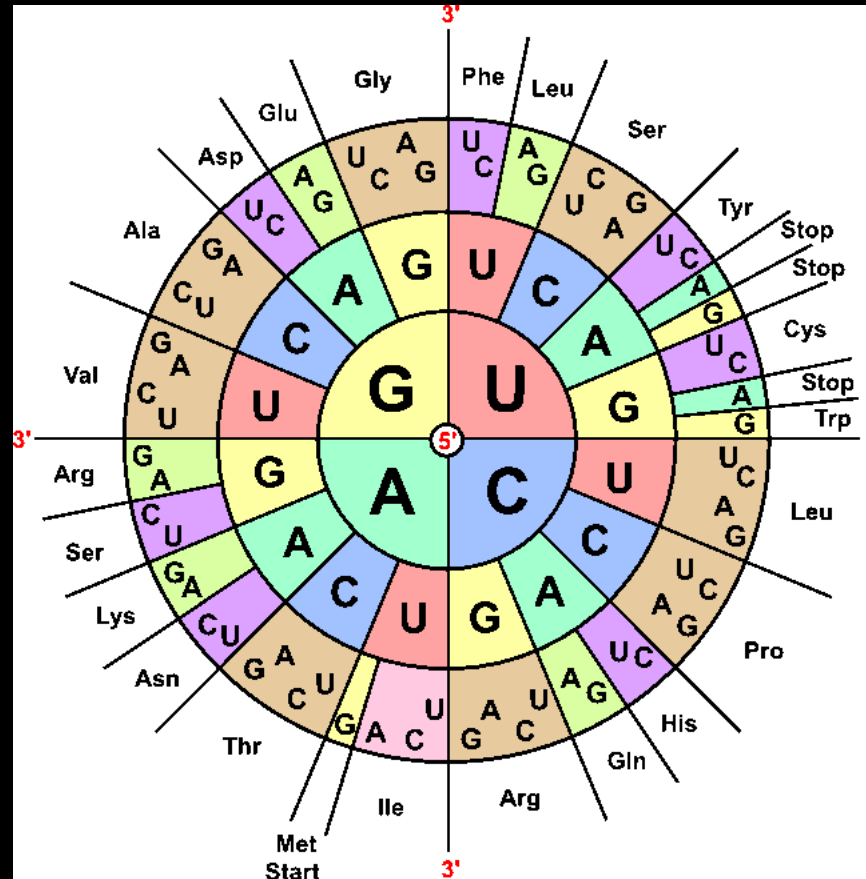
UAG to UAU

Start-gain examples:

CUG to AUG

AGG to AUG

AUC to AUG



Indel/frameshift variants

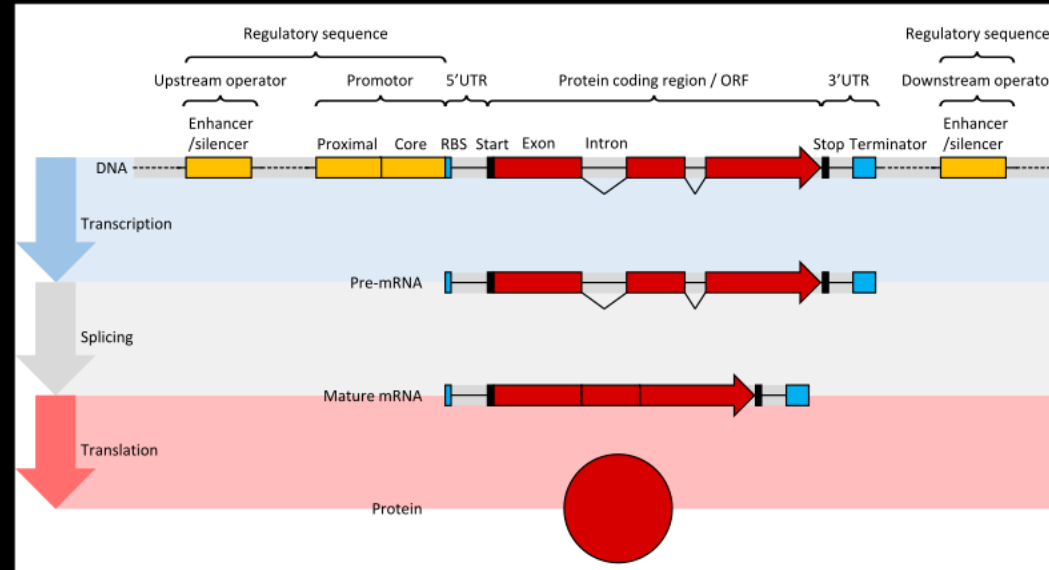
An insertion/deletion (in a gene) whose length isn't divisible by 3 causes a frameshift.

Some examples:

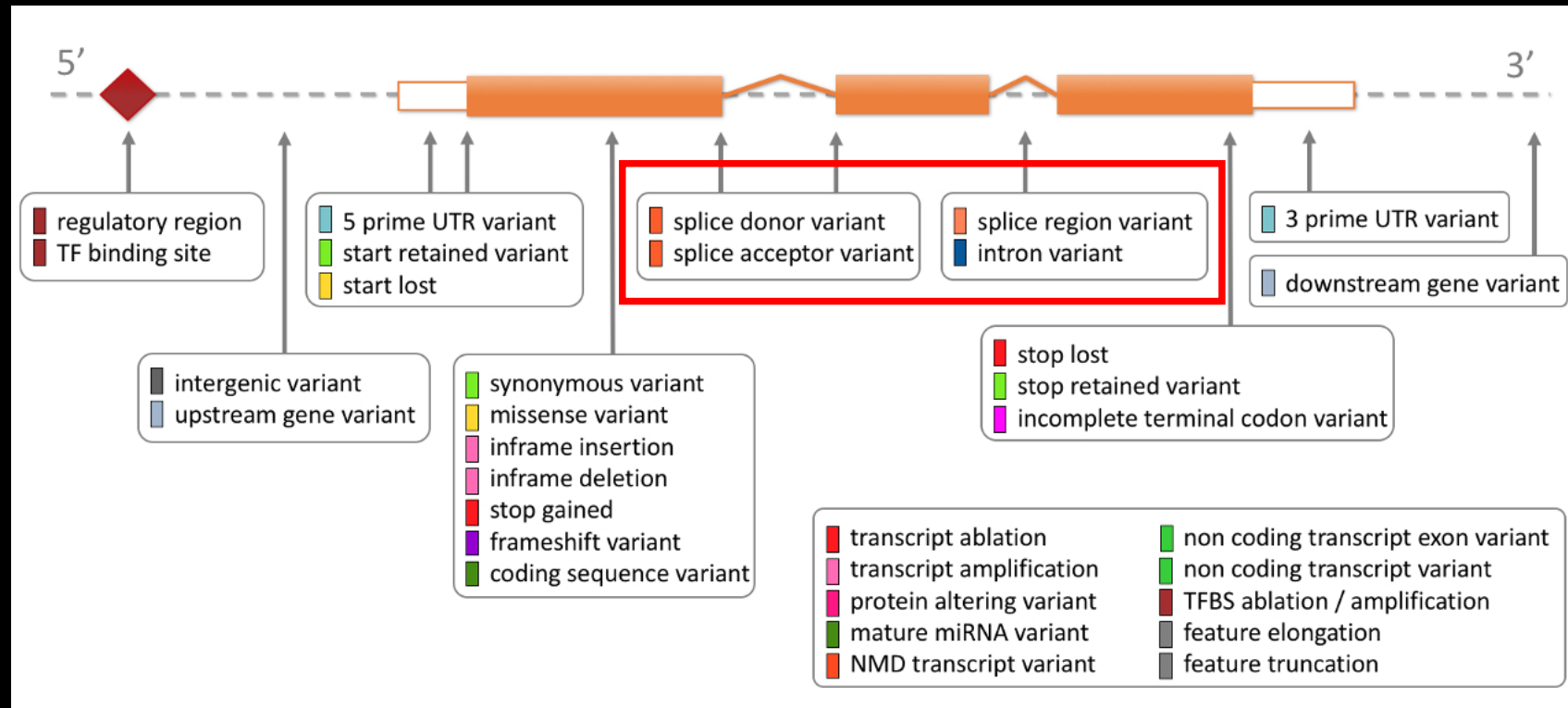
Sequence	Variation	Consequence/description
ATG TTC CCT UTT AGT ...	N/A	Original sequence
ATG G TT CCC TUT TAG T..	1bp insertion	Frameshift. Second and all subsequent amino acids "change."
ATG TTC CCT UAG T... ^TT	2bp deletion	"Frameshift" → Stop gain
ATG TTC UTT AGT ... ^CCT	3bp deletion	One amino acid is deleted; others stay the same.

UTR variation

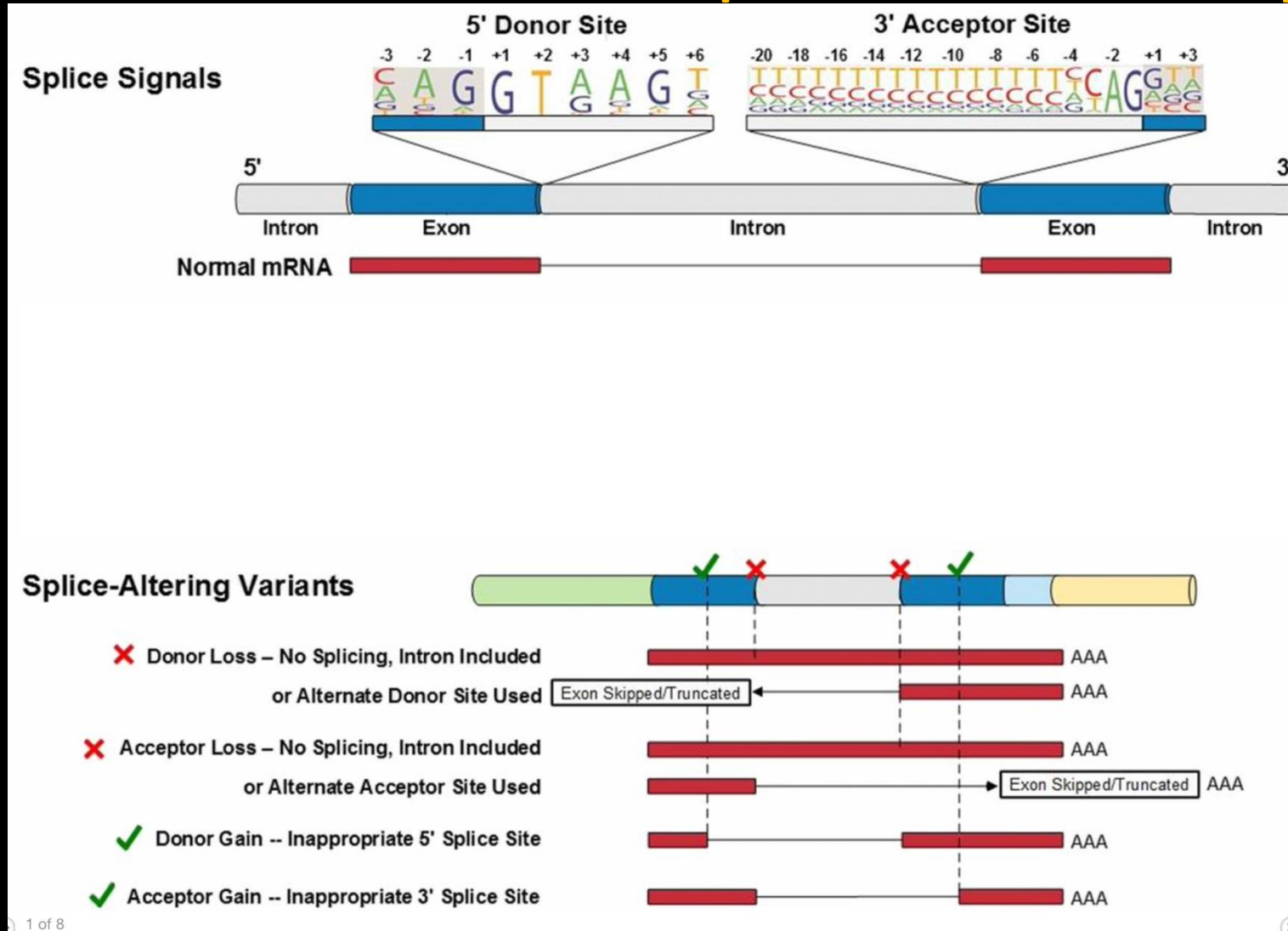
5'-UTR and 3'-UTR (untranlated region) variation may affect gene regulation.



Intron variation



Intron variation – some possible consequences



CADD score

<https://cadd.gs.washington.edu>

For any non-coding variant, is it possible to define a “deleteriousness” score?

CADD = Combined Annotation Dependent Depletion

What is Combined Annotation Dependent Depletion (CADD)?

CADD is a tool for scoring the deleteriousness of single nucleotide variants as well as insertion/deletions variants in the human genome.

While many variant annotation and scoring tools are around, most annotations tend to exploit a single information type (e.g. conservation) and/or are restricted in scope (e.g. to missense changes). Thus, a broadly applicable metric that objectively weights and integrates diverse information is needed. Combined Annotation Dependent Depletion (CADD) is a framework that integrates multiple annotations into one metric by contrasting variants that survived natural selection with simulated mutations.

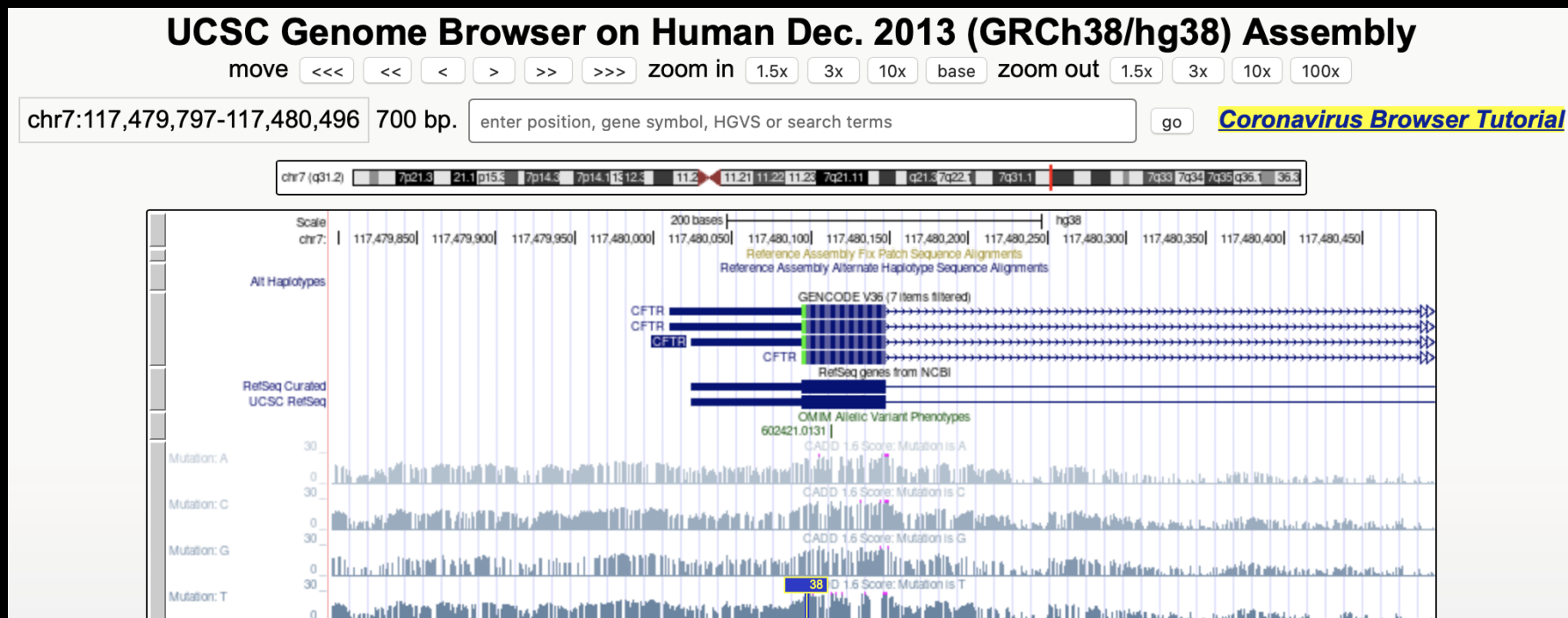
C-scores strongly correlate with allelic diversity, pathogenicity of both coding and non-coding variants, and experimentally measured regulatory effects, and also highly rank causal variants within individual genome sequences. Finally, C-scores of complex trait-associated variants from genome-wide association studies (GWAS) are significantly higher than matched controls and correlate with study sample size, likely reflecting the increased accuracy of larger GWAS.

CADD can quantitatively prioritize functional, deleterious, and disease causal variants across a wide range of functional categories, effect sizes and genetic architectures and can be used to prioritize causal variation in both research and clinical settings.

In addition to this website, CADD has been described in three publications. The most recent manuscript describes CADD-Splice (CADD v1.6), the latest extension of CADD to improve its predictions of splicing effects:

Notes on using scaled vs. unscaled C-scores

Since the raw scores do have relative meaning, one can take a specific group of variants, define the rank for each variant within that group, and then use that value as a "normalized" and now externally comparable unit of analysis. In our case, we scored and ranked all ~8.6 billion SNVs of the GRCh37/hg19 reference and then "PHRED-scaled" those values by expressing the rank in order of magnitude terms rather than the precise rank itself. For example, reference genome single nucleotide variants at the 10th-% of CADD scores are assigned to CADD-10, top 1% to CADD-20, top 0.1% to CADD-30, etc. The results of this transformation are the "scaled" CADD scores.



CADD score

- Access
 - VEP
 - UCSC Genome Browser
 - Upload VCF to CADD website

Clinical significance

We discussed a few kinds of associational evidence that a variant might be involved with a phenotype, for example:

- Being in a gene/regulatory element
- LD with a GWAS association (GWAS Catalog)
- LD with an eQTL

We also talked about various predictive tools (SIFT, PolyPhen, CADD, RegulomeDB score, etc.)

We can also ask what kind of clinical evidence exists for a specific variant (or gene) of interest.

ClinVar

<https://www.ncbi.nlm.nih.gov/clinvar/>

- ClinVar is a database of variants and reports/“assertions” of their relevance to disease/phenotypes
- Entries come from *submissions*
 - Clinical labs
 - Researchers
 - Expert panels
 - Other databases
 - Other sources...
- Only *observed* variation gets into ClinVar
- Assertions about the same variant may be inconsistent; or there may be a consensus; or there may be only a single report

ClinVar

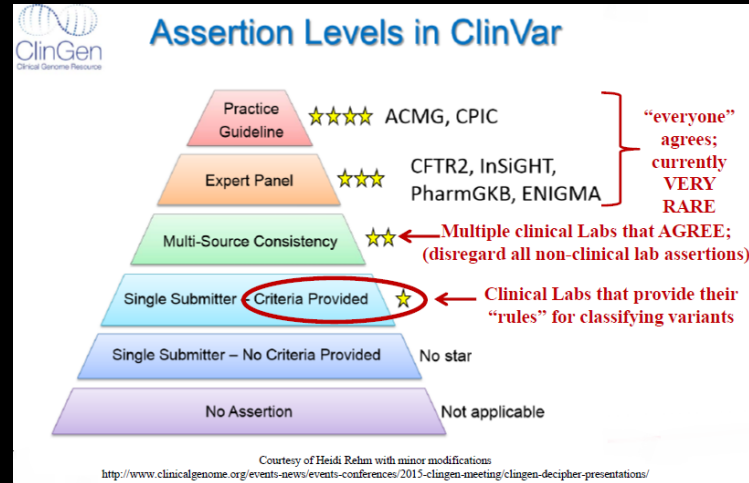


Figure borrowed from:
 Bioinformatics Databases with Clinical Relevance
 Natalie Carter, MS, LCGC
 HUGEN 2070: Bioinformatic Resources for Geneticists
 November 1, 2019

Each variant comes with a “review status” in the form of 0-4 gold stars

Table 1. The review status and assignment of stars

Number of gold stars	Review status	Description
four	practice guideline	practice guideline
three	reviewed by expert panel	reviewed by expert panel
two	criteria provided, multiple submitters, no conflicts	Two or more submitters with assertion criteria and evidence (or a public contact) provided the same interpretation.
one	criteria provided, conflicting interpretations	Multiple submitters provided assertion criteria and evidence (or a public contact) but there are conflicting interpretations. The independent values are enumerated for clinical significance.
one	criteria provided, single submitter	One submitter provided an interpretation with assertion criteria and evidence (or a public contact).
none	no assertion for the individual variant	The allele was not interpreted directly in any submission; it was submitted to ClinVar only as a component of a haplotype or a genotype.
none	no assertion criteria provided	The allele was included in a submission with an interpretation but without assertion criteria and evidence (or a public contact).
none	no assertion provided	The allele was included in a submission that did not provide an interpretation.

ClinVar

Possible searches: variant (HGVS, rsID, gene name, phenotype,)

The screenshot displays the ClinVar website interface. At the top, there is a navigation bar with links for 'NCBI', 'Resources', and 'How To'. A search bar is prominently featured with 'zeb2' entered and a 'Search' button. Below the search bar, a navigation menu includes 'Home', 'About', 'Access', 'Help', 'Submit', 'Statistics', and 'FTP'. A red banner with a warning icon and text states: 'COVID-19 is an emerging, rapidly evolving situation.' Below this banner, a DNA sequence is displayed: 'ACTGATGGTATGGGGCCAAGAGATATATCTCAGGTACGGCTGTCATCACTTAGACCTCACAGGGCTGGGCATAAAAGTCAGGGCAGAGCCATGGTGCATCTGACTCCTGAGGAGAAGTGCAGGTTGGTATCAAGGTTACAAGACAGGTGGCACTGACTCTCTCTGCCTATTGGTCTAT'. To the right of the sequence, the ClinVar logo and a brief description are shown. Below the sequence, there are three columns of links: 'Using ClinVar' (including 'About ClinVar', 'Data Dictionary', 'Downloads/FTP site', 'FAQ', 'Contact Us', and 'Factsheet'), 'Tools' (including 'ACMG Recommendations for Reporting of Incidental Findings', 'ClinVar Submission Portal', 'Submissions', 'Variation Viewer', 'Clinical Remapping - Between assemblies and RefSeqGenes', and 'RefSeqGene/LRG'), and 'Related Sites' (including 'ClinGen', 'GeneReviews @', 'GTR @', 'MedGen', 'OMIM @', and 'Variation'). At the bottom, there is a 'Submitter highlights' section and a 'Disclaimer' section.

NCBI Resources How To Sign in to NCBI

ClinVar ClinVar zeb2 Search Advanced Help

Home About Access Help Submit Statistics FTP

COVID-19 is an emerging, rapidly evolving situation.
[Public health information \(CDC\)](#) | [Research information \(NIH\)](#) | [SARS-CoV-2 data \(NCBI\)](#) | [Prevention and treatment information \(HHS\)](#)

ACTGATGGTATGGGGCCAAGAGATATATCTCAGGTACGGCTGTCATCACTTAGACCTCACAGGGCTGGGCATAAAAGTCAGGGCAGAGCCATGGTGCATCTGACTCCTGAGGAGAAGTGCAGGTTGGTATCAAGGTTACAAGACAGGTGGCACTGACTCTCTCTGCCTATTGGTCTAT

ClinVar
ClinVar aggregates information about genomic variation and its relationship to human health.

Using ClinVar
[About ClinVar](#)
[Data Dictionary](#)
[Downloads/FTP site](#)
[FAQ](#)
[Contact Us](#)
[Factsheet](#)

Tools
[ACMG Recommendations for Reporting of Incidental Findings](#)
[ClinVar Submission Portal](#)
[Submissions](#)
[Variation Viewer](#)
[Clinical Remapping - Between assemblies and RefSeqGenes](#)
[RefSeqGene/LRG](#)

Related Sites
[ClinGen](#)
[GeneReviews @](#)
[GTR @](#)
[MedGen](#)
[OMIM @](#)
[Variation](#)

Submitter highlights
We gratefully acknowledge those who have submitted data and provided advice during the development of ClinVar.
Follow us on [Twitter](#) to receive announcements of the release of new datasets.
Want to learn more about who submits to ClinVar?

- [Read information about groups that submit to ClinVar](#)
- [See the list of submitters with the number of records each has submitted](#)
- [View a world map of ClinVar submitters](#)

Disclaimer
The information on this website is not intended for direct diagnostic use or medical decision-making without review by a genetics professional. Individuals should not change their health behavior solely on the basis of information contained on this website. NIH does not independently verify the submitted information. If you have questions about the information contained on this website, please see a [health care professional](#). [More information about NCBI's disclaimer policy is available.](#)

Search ClinVar for a gene

Result: all variants in that gene

NCBI

Resources

How To

Sign in to NCBI

ClinVar

Search

Create alert

Advanced

Help

Home

About

Access

Help

Submit

Statistics

FTP

COVID-19 is an emerging, rapidly evolving situation.

[Public health information \(CDC\)](#) | [Research information \(NIH\)](#) | [SARS-CoV-2 data \(NCBI\)](#) | [Prevention and treatment information \(HHS\)](#)

Clinical significance

Conflicting interpretations (38)

Benign (67)

Likely benign (168)

Uncertain significance (168)

Likely pathogenic (25)

Pathogenic (202)

Risk factor (0)

Molecular consequence

Frameshift (107)

Missense (236)

Nonsense (62)

Splice site (8)

ncRNA (20)

Near gene (0)

UTR (26)

Variation type

Deletion (117)

Duplication (44)

Indel (8)

Insertion (50)

Single nucleotide (450)

Variant length

Less than 51 bp (601)

Between 51 and 1000 bp (3)

Between 1 and 50 kb (0)

Between 50 and 500 kb (6)

Between 500 kb and 1 Mb (3)

Between 1 and 5 Mb (2)

Greater than 5 Mb (4)

Review status

Practice guideline (0)

Expert panel (0)

Multiple submitters (91)

Single submitter (440)

At least one star (569)

Tabular

100 per page

Sort by Location

Download:

Showing for results for variants in the **zeb2** gene. [Search instead for all ClinVar records that mention zeb2](#)

Search results

Items: 1 to 100 of 637

<< First < Prev Page 1 of 7 Next > Last >>

	Variation Location	Gene(s)	Protein change	Condition(s)	Clinical significance (Last reviewed)	Review status	Accession
☐	GRCh38/hg38 2q22.1-22.3(chr2:13693735-8-146681810)x1 GRCh37: Chr2:137694928-147439378 GRCh38: Chr2:136937358-146681810	HNMT , KYNJ , ZEB2 , NXPH2 , LRP1B , ARHGAP15 , GTD1C1 , THSD7B , SPOPL , TEX41 , ZEB2-AS1 , LOC101928273 , LOC101928386 , LINC01412 , MIR7157 , LINC01832 , LINC01853 , ARHGAP15-AS1 , LOC105373656 , LINC01966 , LOC107228318 , LOC110120671 , LOC110120706 , LOC110120715 , ... more		See cases	Pathogenic (May 12, 2011)	no assertion criteria provided	VCV000148432
☐	GRCh38/hg38 2q22.1-22.3(chr2:14018652-1-146528244)x1 GRCh37: Chr2:140944090-147285812 GRCh38: Chr2:140186521-146528244	KYNJ , ZEB2 , LRP1B , ARHGAP15 , GTD1C1 , TEX41 , ZEB2-AS1 , LOC101928386 , LINC01412 , MIR7157 , ARHGAP15-AS1 , LOC105373656 , LINC01966 , LOC107228318 , LOC110120671 , LOC110120706 , LOC110120715 , LOC110120731 , LOC110120740 , LOC110121209 , LOC111556162 , ... more		See cases	Pathogenic (Aug 12, 2011)	criteria provided, single submitter	VCV000057333
☐	GRCh38/hg38 2q22.2-22.3(chr2:14166653-7-147845662)x1 GRCh37: Chr2:142424106-148603231 GRCh38: Chr2:141666537-147845662	ACVR2A , KYNJ , ZEB2 , LRP1B , ARHGAP15 , GTD1C1 , TEX41 , ZEB2-AS1 , LOC101928386 , LINC01412 , ARHGAP15-AS1 , LOC105373656 , LINC01966 , LINC01911 , LOC107228318 , LOC110120671 , LOC110120706 , LOC110120715 , LOC110120731 , LOC110120740 , LOC110121209 , LOC111556162 , ... more		See cases	Pathogenic (Aug 12, 2011)	criteria provided, single submitter	VCV000060212
☐	GRCh38/hg38 2q22.2-22.3(chr2:14329407-9-147713830)x1 GRCh37: Chr2:144051648-148471399 GRCh38: Chr2:143294079-147713830	ZEB2 , ARHGAP15 , GTD1C1 , TEX41 , ZEB2-AS1 , LOC101928386 , LINC01412 , ARHGAP15-AS1 , LOC105373656 , LINC01966 , LINC01911 , LOC107228318 , LOC110120671 , LOC110120706 , LOC110120715 , LOC110120731 , LOC110120740 , LOC110121209 , LOC111721705 , LOC112806051 , ... more		See cases	Pathogenic (May 3, 2012)	no assertion criteria provided	VCV000150244
☐	GRCh38/hg38 2q22.3(chr2:143873711-144513307)x1 GRCh37: Chr2:144631279-145270874	ZEB2 , GTD1C1 , LOC101928386 , LOC110120671 , LOC111721705 , LOC112806051		See cases	Pathogenic (Aug 12, 2011)	criteria provided, single	VCV000060213

Search ClinVar for a gene

Use the panel at the left to filter

- Select “Pathogenic” variants, e.g.
- Select only “large” variants > 500kb in length, e.g.
- Select only genes with multiple submitters, e.g.

Clinical significance
Conflicting interpretations (38)
Benign (67)
Likely benign (168)
Uncertain significance (168)
Likely pathogenic (25)
Pathogenic (202)
Risk factor (0)

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Between 50 and 500 kb (6)
Between 500 kb and 1 Mb (3)
Between 1 and 5 Mb (2)
Greater than 5 Mb (4)

Review status
Practice guideline (0)
Expert panel (0)
Multiple submitters (91)
Single submitter (440)
At least one star (569)
Conflicting interpretations (38)

Allele origin
Germline (557)
De novo (30)
Somatic (1)

Method type
Research (9)
Literature only (17)
Clinical testing (581)

Variant-gene relationship
Single gene (596)

Tabular ▾ 100 per page ▾ Sort by Location ▾

 Showing for results for variants in the **zeb2** gene

Search results

Items: 1 to 100 of 637

Variation Location	
☐	GRCh38/hg38 2q22.1-22.3(chr2:136937358-146681810)x1 <i>GRCh37:</i> Chr2:137694928-147439378 <i>GRCh38:</i> Chr2:136937358-146681810
1.	
☐	GRCh38/hg38 2q22.1-22.3(chr2:140186521-146528244)x1 <i>GRCh37:</i> Chr2:140944090-147285812 <i>GRCh38:</i> Chr2:140186521-146528244
2.	
☐	GRCh38/hg38 2q22.2-22.3(chr2:141666537-147845662)x1 <i>GRCh37:</i> Chr2:142424106-148603231 <i>GRCh38:</i> Chr2:141666537-147845662
3.	
☐	GRCh38/hg38 2q22.2-22.3(chr2:143294079-147713830)x1 <i>GRCh37:</i> Chr2:144051648-148471399 <i>GRCh38:</i> Chr2:143294079-147713830
4.	
☐	GRCh38/hg38 2q22.3(chr2:143873711-144513307)x1 <i>GRCh37:</i> Chr2:144631279-145270874 <i>GRCh38:</i> Chr2:143873711-144513307
5.	
☐	GRCh38/hg38 2q22.3-24.1(chr2:143900149-158321624)x3 <i>GRCh37:</i> Chr2:144657717-159178136 <i>GRCh38:</i> Chr2:143900149-158321624
6.	
☐	GRCh38/hg38 2q22.3(chr2:143900149-144668138)x3 <i>GRCh37:</i> Chr2:144657717-145425705 <i>GRCh38:</i> Chr2:143900149-144668138
7.	
☐	GRCh38/hg38 2q22.3(chr2:143988786-144668138)x3

Search ClinVar for a variant

For example, search by HGVS name
("NM_004333.4:c.1799T>A")

This leads directly to the variant's page,
since there's only 1 search result

Can also enter queries like:

- 1799T>A BRAF
- 1799T>A
- rs113488022
- NP_004324.2:p.Val600Glu
- P.Val600Glu
- V600E
- (You may get multiple results)

NM_001374258.1(BRAF):c.1919T>A (p.Val640Glu)

Cite this record

Interpretation:Pathogenic, drug response, other

Review status:☆☆☆ criteria provided, multiple submitters, no conflicts

Submissions:34 (Most recent: Nov 26, 2020)

Last evaluated:Jul 11, 2014

Accession:VCV000013961.13

Variation ID:13961

Description:single nucleotide variant

Variant details

Conditions

Gene(s)

NM_001374258.1(BRAF):c.1919T>A (p.Val640Glu)

Allele ID:29000

Variant type:single nucleotide variant

Variant length:1 bp

Cytogenetic location:7q34

Genomic location:7: 140753336 (GRCh38) GRCh38 UCSC
7: 140453136 (GRCh37) GRCh37 UCSC
7: 140099605 (NCBI36) NCBI36 UCSC

HGVS:

Nucleotide	Protein	Molecular consequence
NC_000007.12:g.140099605A>T		
NC_000007.13:g.140453136A>T		
NC_000007.14:g.140753336A>T		

... more HGVS

Protein change:V600E, V512E, V578E, V603E, V548E, V566E, V563E, V640E

Other names:-

Canonical SPDI: NC_000007.14:140753335:A:T

Functional consequence:-

Global minor allele frequency (GMAF):-

Allele frequency:Exome Aggregation Consortium (ExAC) 0.00002
The Genome Aggregation Database (gnomAD), exomes 0.00000

Links:

ClinGen: CA123643

Genetic Testing Registry (GTR): GTR000522729

Genetic Testing Registry (GTR): GTR000560765

Genetic Testing Registry (GTR): GTR000575664

Genetic Testing Registry (GTR): GTR000575667

Genetic Testing Registry (GTR): GTR000575672

Genetic Testing Registry (GTR): GTR000575677

UniProtKB: P15056#VAR_018629

OMIM: 164757.0001

dbSNP: rs113488022

Searching ClinVar for a variant (rs113488022)

	Variation Location	Gene(s)	Protein change	Condition(s)	Clinical significance (Last reviewed)	Review status	Accession
☐ 1.	NM_004333.6(BRAF):c.1799T>C (p.Val600Ala) GRCh37: Chr7:140453136 GRCh38: Chr7:140753336	BRAF	V600A	Melanoma	Likely pathogenic (Jul 14, 2015)	no assertion criteria provided	VCV000376288
☐ 2.	NM_004333.6(BRAF):c.1799T>G (p.Val600Gly) GRCh37: Chr7:140453136 GRCh38: Chr7:140753336	BRAF	V600G	Melanoma, Cardio-facio-cutaneous syndrome, Rasopathy	Pathogenic (Jun 25, 2020)	reviewed by expert panel FDA Recognized Database	VCV000040389
☐ 3.	NM_001374258.1(BRAF):c.1919T>A (p.Val640Glu) GRCh37: Chr7:140453136 GRCh38: Chr7:140753336	BRAF	V600E, V512E, V578E, V603E, V548E, V566E, V563E, V640E	Cancer, Melanoma, Trametinib-Dabrafenib Response, Vemurafenib-Cobimetinib Response, Neoplasm of brain, Colonic neoplasm, Papillary renal cell carcinoma, sporadic, Neoplasm, Cystic epithelial invagination containing papillae lined by columnar epithelium, Lung adenocarcinoma, Neoplasm of ovaryAstrocytoma, low-grade, somatic, Squamous cell carcinoma of the head and neck, Glioblastoma, Nephroblastoma, Gastrointestinal stromal tumor, Multiple myeloma, not provided, Brainstem glioma, Germ cell tumor, nonseminomatous, Colorectal cancer, Malignant melanoma of skin, Cerebral arteriovenous malformation, Lung carcinoma, Papillary thyroid carcinoma, Neoplasm of the large intestine, Carcinoma of colon, Cardio-facio-cutaneous syndrome, Non-small cell lung cancer, ...see more	Pathogenic, drug response, other (Jul 11, 2014)	criteria provided, multiple submitters, no conflicts	VCV000013961

Searching ClinVar for a region: Using the Variation Viewer

Using ClinVar

[About ClinVar](#)

[Data Dictionary](#)

[Downloads/FTP site](#)

[FAQ](#)

[Contact Us](#)

[Factsheet](#)

Tools

[ACMG Recommendations for Reporting of Incidental Findings](#)

[ClinVar Submission Portal](#)

[Submissions](#)

[Variation Viewer](#)

[Clinical Remapping - Between assemblies and RefSeqGenes](#)

[RefSeqGene/LRG](#)

Related Sites

[ClinGen](#)

[GeneReviews ®](#)

[GTR ®](#)

[MedGen](#)

[OMIM ®](#)

[Variation](#)

Variation Viewer

Search: ZEB2 gene

NIH U.S. National Library of Medicine NCBI National Center for Biotechnology Information

Log in

Variation Viewer

Homo sapiens (human)

Assembly: GRCh38.p12 (GCF_000001405.38) • Chr 2 (NC_000002.12)

Search assembly: zeb2

Region: ZEB2 Transcript: NM_014795.4

NC_000002.12: 144,370,477 - 144,533,723

Genes, NCBI Homo sapiens Annotation Release 109.20210226

ClinVar Short Variations based on dbSNP Build 150 (Homo sapiens Annotation Release 108)

dbVar Clinical Structural Variants (nstd102)

dbVar Non-Pathogenic Clinical Structural Variants (subset of nstd102)

dbVar Pathogenic Clinical Structural Variants (subset of nstd102)

Clinical, dbSNP b154 v2

Live RefSNPs, dbSNP b154 v2

NC_000002.12: 144M.145M (163,247 nt)

Variation Data

Filter by

Source database

dbSNP (37307)

dbVar (432)

In ClinVar

Yes (592)

No (37147)

Most severe clinical significance

pathogenic (187)

pathogenic-likely-pathogenic (1)

likely-pathogenic (15)

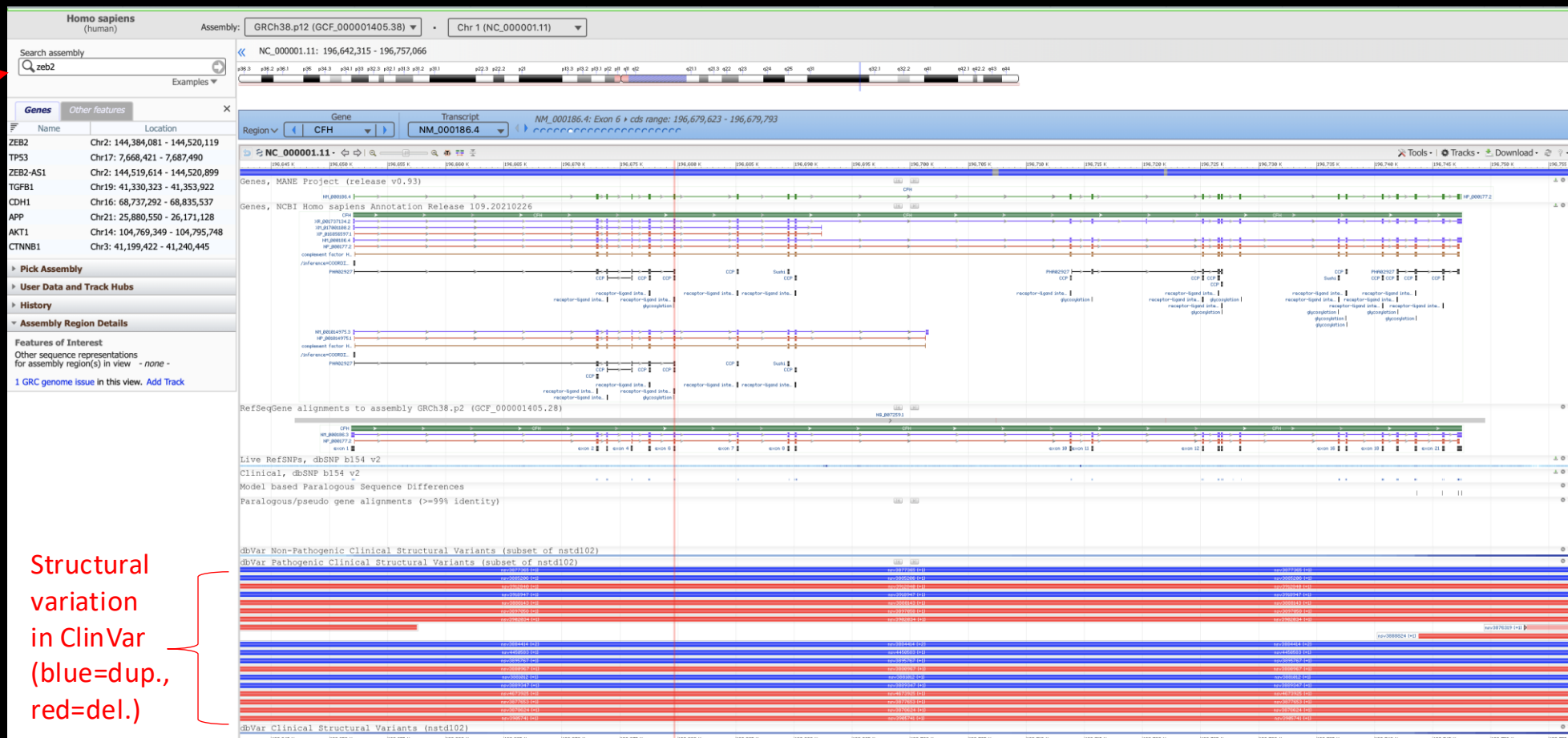
drug-response (1)

confers sensitivity (0)

Variant ID	Location	Variant type	Gene	Molecular consequences	Most severe clinical significance	1000G MAF	GO-ESP MAF	ExAC MAF	Publications
nsv2776363	12,771 - 241,841,232	copy number variation	H2ACP2 and 3864 more		Pathogenic				1
nsv3882615	12,771 - 241,841,232	copy number variation	H2ACP2 and 3864 more		Pathogenic				
nsv429565	12,784 - 242,147,305	copy number variation	H2ACP2 and 3877 more						1
nsv3885544	14,238 - 242,106,609	copy number variation	H2ACP2 and 3875 more		Pathogenic				
nsv3874648	15,672 - 242,157,305	copy number variation	H2ACP2 and 3877 more		Pathogenic				
nsv535502	36,551 - 242,063,877	copy number variation	H2ACP2 and 3874 more						1
nsv1004728	36,551 - 242,063,877	copy number variation	H2ACP2 and 3874 more						1
esv3766985	1,155,459 - 238,670,674	tandem duplication	H2ACP2 and 3764 more						2
esv3812160	5,383,622 - 232,981,193	tandem duplication	H2ACP2 and 3613 more						2
esv3768537	13,043,053 - 177,979,093	tandem duplication	LRP1B and 2627 more						2
esv3804955	17,706,767 - 166,039,161	tandem duplication	LRP1B and 2395 more						2

Variation Viewer

Search: ZEB2 gene



Variation Viewer

Zoom in/out

Zoom in all the way to base-pair level

Zoom into an exon of your choice

Edit tracks

Variants in ClinVar

The screenshot displays the UCSC Genome Browser Variation Viewer for the ZEB2 gene (NC_000002.12). The top track shows the gene structure with exons and introns. Below the gene track is a zoom slider and a search bar. The main track shows the gene structure with exons and introns. The ClinVar Short Variations track shows variants based on dbSNP Build 150 (Homo sapiens Annotation Release 108). The dbVar Clinical Structural Variants (nstd102) track shows structural variants. The dbVar Pathogenic Clinical Structural Variants (subset of nstd102) track shows pathogenic structural variants. The Clinical, dbSNP b154 v2 track shows clinical SNPs. The Live RefSNPs, dbSNP b154 v2 track shows live reference SNPs.

Track	Content
Genes, NCBI Homo sapiens Annotation Release 109.20210226	Gene structure of ZEB2 (+5)
ClinVar Short Variations based on dbSNP Build 150 (Homo sapiens Annotation Release 108)	9 variants (green and purple icons)
dbVar Clinical Structural Variants (nstd102)	1 variant (purple icon)
dbVar Pathogenic Clinical Structural Variants (subset of nstd102)	0 variants
Clinical, dbSNP b154 v2	0 variants
Live RefSNPs, dbSNP b154 v2	0 variants

Variation Viewer

Below the viewer window is a table of all the ClinVar variants currently in the window. The table updates automatically as the window moves.

ClinVar Short Variations based on dbSNP Build 150 (Homo sapiens Annotation Release 108)

dbVar Clinical Structural Variants (nstd102)

dbVar Pathogenic Clinical Structural Variants (subset of nstd102)

Clinical, dbSNP b154 v2

Live RefSNPs, dbSNP b154 v2

44,370 K

144,380 K

144,390 K

144,400 K

144,410 K

144,420 K

144,430 K

144,440 K

144,450 K

144,460 K

144,470 K

144,480 K

144,490 K

144,500 K

144,510 K

144,520 K

144,530 K

NC_000002.12: 144M..145M (163,247 nt)

Tracks shown: 7/665

Variation Data

Download

edit columns

Items 1 - 30 of 37,739 << First < Prev Page 1 of 1,258 Next > Last >>

	Variant ID	Location	Variant type	Gene	Molecular consequences	Most severe clinical significance	1000G MAF	GO-ESP MAF	ExAC MAF	Publications		
Source database ☐ dbSNP (37307) ☐ dbVar (432)	nsv2776363	12,771 - 241,841,232	copy number variation	H2ACP2 and 3864 more		Pathogenic				1		
	nsv3882615	12,771 - 241,841,232	copy number variation	H2ACP2 and 3864 more		Pathogenic						
	nsv429565	12,784 - 242,147,305	copy number variation	H2ACP2 and 3877 more						1		
	nsv3885544	14,238 - 242,106,609	copy number variation	H2ACP2 and 3875 more		Pathogenic						
	nsv3874648	15,672 - 242,157,305	copy number variation	H2ACP2 and 3877 more		Pathogenic						
In ClinVar ☐ Yes (592) ☐ No (37147)	nsv535502	36,551 - 242,063,877	copy number variation	H2ACP2 and 3874 more						1		
	nsv1004728	36,551 - 242,063,877	copy number variation	H2ACP2 and 3874 more						1		
	esv3766985	1,155,459 - 238,670,674	tandem duplication	H2ACP2 and 3764 more						2		
	esv3812160	5,383,622 - 232,981,193	tandem duplication	H2ACP2 and 3613 more						2		
	esv3768537	13,043,053 - 177,979,093	tandem duplication	LRP1B and 2627 more						2		
	esv3804065	17,706,767 - 166,039,151	tandem duplication	LRP1B and 2395 more						2		
	nsv435981	21,986,284 - 174,843,193	inversion	LRP1B and 2458 more						1		
	esv4002915	22,127,393 - 161,340,084	inversion	DPP10 and 2279 more						1		
	nsv3167831	32,866,630 - 206,122,980	inversion	LRP1B and 2699 more						1		
	nsv3168311	32,866,779 - 202,040,363	copy number variation	LRP1B and 2631 more						1		
Most severe clinical significance ☐ pathogenic (187) ☐ pathogenic-likely-pathogenic (1) ☐ likely-pathogenic (15) ☐ drug-response (1) ☐ confers sensitivity (0) More...	nsv3167784	32,915,920 - 155,133,453	inversion	LOC730100 and 1979 more						1		
	nsv3169185	32,916,345 - 239,742,818	inversion	H2ACP2 and 3294 more						1		
	esv3761036	34,562,120 - 237,470,652	tandem duplication	H3P9 and 3227 more						2		
	esv3798086	36,109,730 - 186,650,334	tandem duplication	LRP1B and 2379 more						2		
	nsv3923060	40,379,902 - 146,145,196	inversion	LOC730100 and 1774 more		Likely-Pathogenic				1		
	esv3783136	43,386,294 - 172,680,115	copy number variation	LOC730100 and 2055 more						2		
	esv3812928	46,625,879 - 147,904,876	inversion	LOC730100 and 1688 more						3		
	nsv2772356	50,659,294 - 242,160,331	copy number variation	H3P9 and 3107 more		Pathogenic				1		
	nsv3910630	50,659,294 - 242,160,331	copy number variation	H3P9 and 3107 more		Pathogenic						
	esv3808247	51,053,652 - 210,158,776	tandem duplication	LRP1B and 2513 more						3		
Variant type ☐ single nucleotide variant (33667) ☐ copy number variation (230) ☐ deletion (658) ☐ insertion (258) ☐ microsatellite (0) More...	esv3793473	54,214,675 - 189,901,887	copy number variation	DPP10 and 2157 more						2		
	esv3798687	63,673,400 - 175,314,504	tandem duplication	DPP10 and 1825 more						2		
	esv3774701	63,673,412 - 175,314,491	copy number variation	DPP10 and 1825 more						2		
	esv3776712	64,104,253 - 237,083,295	copy number variation	LRP1B and 2806 more						2		
	esv3801224	76,499,414 - 186,650,375	copy number variation	DPP10 and 1702 more						2		
	Molecular consequence ☐ missense variant (630) ☐ nonsense (stop gained) (54) ☐ stop lost (1) ☐ inframe variant (0) ☐ frameshift variant (94) More...											
1000 Genomes MAF ☐ not specified (33890) ☐ < 0.005 (3362) ☐ 0.005 - 0.01 (115) ☐ 0.01 - 0.05 (166) ☐ >= 0.05 (206)												
GO_ESP MAF												

Items 1 - 30 of 37,739 << First < Prev Page 1 of 1,258 Next > Last >>

Searching ClinVar for traits

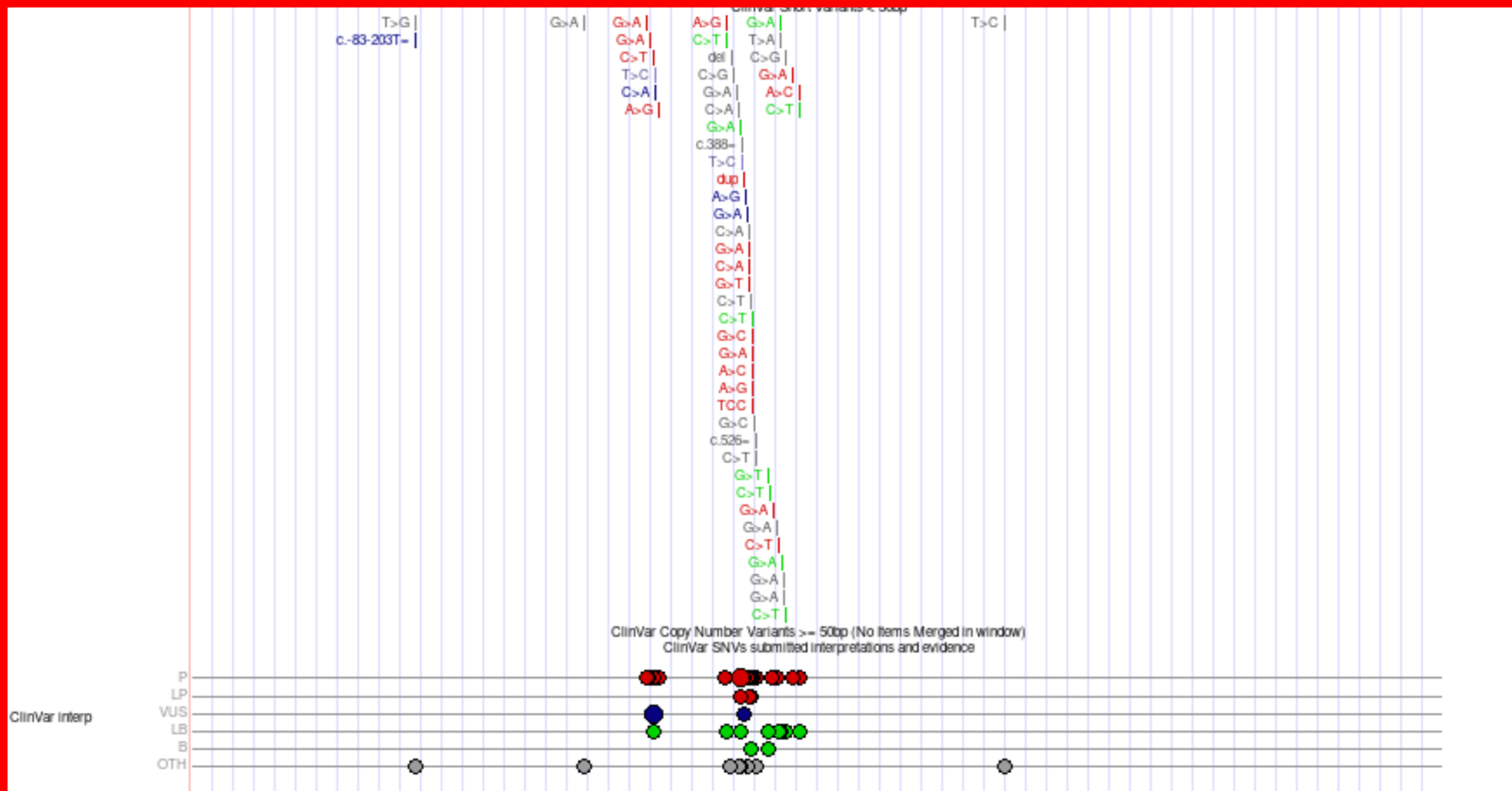
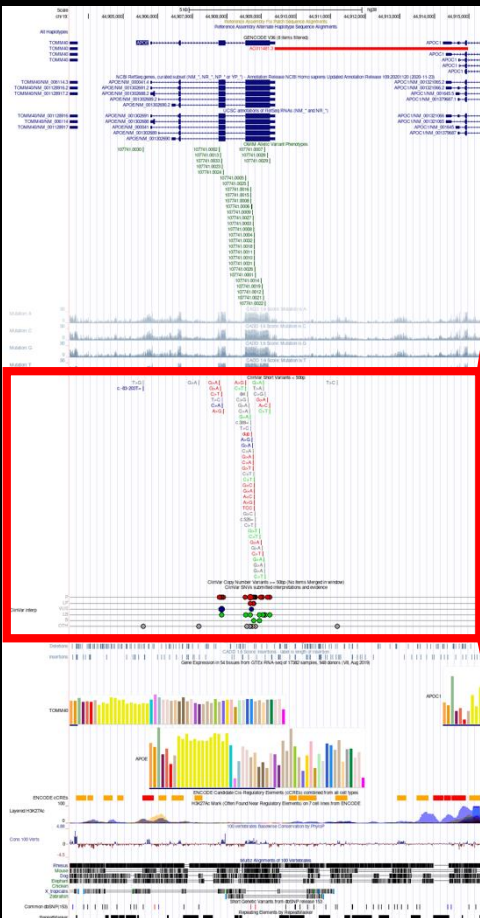
I searched “cystic fibrosis.”

Note that CF isn’t mentioned in the Condition(s).

Your search term may show up somewhere in the individual variants’ entries.

Search results							
Items: 1 to 100 of 4790				<< First < Prev Page 1 of 48 Next > Last >>			
	Variation Location	Gene(s)	Protein change	Condition(s)	Clinical significance (Last reviewed)	Review status	Accession
☐ 1.	NM_015102.5(NPHP4):c.*314T>G GRCh37: Chr1:5923011 GRCh38: Chr1:5862951	NPHP4		Nephronophthisis 4, Senior-Loken syndrome 4	Benign (Jan 13, 2018)	criteria provided, single submitter	VCV000297781
☐ 2.	NM_015102.5(NPHP4):c.*285G>A GRCh37: Chr1:5923040 GRCh38: Chr1:5862980	NPHP4		Nephronophthisis 4, Senior-Loken syndrome 4	Uncertain significance (Jan 13, 2018)	criteria provided, single submitter	VCV000876727
☐ 3.	NM_015102.5(NPHP4):c.*128A>C GRCh37: Chr1:5923197 GRCh38: Chr1:5863137	NPHP4		Senior-Loken syndrome 4, Nephronophthisis 4	Benign (Jan 13, 2018)	criteria provided, single submitter	VCV000297782
☐ 4.	NM_015102.5(NPHP4):c.*97T>A GRCh37: Chr1:5923228 GRCh38: Chr1:5863168	NPHP4		Senior-Loken syndrome 4, Nephronophthisis 4	Uncertain significance (Jan 12, 2018)	criteria provided, single submitter	VCV000873895
☐ 5.	NM_015102.5(NPHP4):c.*87C>G GRCh37: Chr1:5923238 GRCh38: Chr1:5863178	NPHP4		Senior-Loken syndrome 4, Nephronophthisis 4	Uncertain significance (Jan 13, 2018)	criteria provided, single submitter	VCV000873896
☐ 6.	NM_015102.5(NPHP4):c.*26C>T GRCh37: Chr1:5923299 GRCh38: Chr1:5863239	NPHP4		Nephronophthisis 4, Senior-Loken syndrome 4	Uncertain significance (Jan 13, 2018)	criteria provided, single submitter	VCV000297783
☐ 7.	NM_015102.5(NPHP4):c.4267G>A (p.Val1423Ile) GRCh37: Chr1:5923299 GRCh38: Chr1:5863239	NPHP4	V1423I, V911I, V910I	Nephronophthisis	Uncertain significance (Dec 20, 2018)	criteria provided, single submitter	VCV000843518

ClinVar track in UCSC Genome Browser



ClinVar tracks in UCSC Genome Browser

Here the short ClinVar variants are squished,
and the structural variants are shown.

The blue bars represent
“pathogenic” gains in copy number.



ClinVar annotation in VEP

Variant Effect Predictor results

Job details

Summary statistics

Results preview

Navigation (per variant)

Page: 1 of 111 | Show: 1 5 10 50 All variants

Filters

Uploaded variant is defined Add

Download

New job

All: VCF VEP TXT

BioMart: Variants Genes

Show/hide columns (17 hidden)

Uploaded variant	Location	Allele	Consequence	Symbol	Gene	Feature type	Feature	Biotype	Exon	HGVSc	cDNA position	Existing variant	Feature strand	MANE SELECT	Transcript support level	APPRIS	AF	Clinical significance	Pubmed	Motif
rs115960004	1:26795430-26795430	A	intron_variant	PIGV	ENSG00000060642	Transcript	ENST00000078527.9	protein_coding	-	ENST00000078527.9:c.1200+196C>A	-	rs115960004	1	-	1					
rs115960004	1:26795430-26795430	A	intron_variant	PIGV	ENSG00000060642	Transcript	ENST000000374145.6	protein_coding	-	ENST000000374145.6:c.1200+196C>A	-	rs115960004	1	-	2					
rs115960004	1:26795430-26795430	A	intron_variant	PIGV	ENSG00000060642	Transcript	ENST000000431541.6	protein_coding	-	ENST000000431541.6:c.1180+216C>A	-	rs115960004	1	-	2					
rs115960004	1:26795430-26795430	A	intron_variant	PIGV	ENSG00000060642	Transcript	ENST000000455364.2	protein_coding	-	ENST000000455364.2:c.1200+196C>A	-	rs115960004	1	-	3					
rs115960004	1:26795430-26795430	A	intron_variant, NMD transcript variant	PIGV	ENSG00000060642	Transcript	ENST000000472757.5	nonsense_mediated_decay	-	ENST000000472757.5:c.*379+196C>A	-	rs115960004	1	-	5					
rs115960004	1:26795430-26795430	A	intron_variant	PIGV	ENSG00000060642	Transcript	ENST000000674202.1	protein_coding	-	ENST000000674202.1:c.1200+196C>A	-	rs115960004	1	NM_017837.4	-					
rs115960004	1:26795430-26795430	A	intron_variant	PIGV	ENSG00000060642	Transcript	ENST000000674222.1	protein_coding	-	ENST000000674222.1:c.1200+196C>A	-	rs115960004	1	-	-					
rs115960004	1:26795430-26795430	A	intron_variant	PIGV	ENSG00000060642	Transcript	ENST000000674273.1	protein_coding	-	ENST000000674273.1:c.1200+196C>A	-	rs115960004	1	-	-					
rs115960004	1:26795430-26795430	A	intron_variant, non_coding_transcript_variant	PIGV	ENSG00000060642	Transcript	ENST000000674317.1	processed_transcript	-	ENST000000674317.1:n.1505+196C>A	-	rs115960004	1	-	-					
rs115960004	1:26795430-26795430	A	intron_variant	PIGV	ENSG00000060642	Transcript	ENST000000674335.1	protein_coding	-	ENST000000674335.1:c.819+196C>A	-	rs115960004	1	-	-					
rs71636780	1:26787956-26787956	A	upstream_gene_variant	PIGV	ENSG00000060642	Transcript	ENST00000078527.9	protein_coding	-	-	-	rs71636780	1	-	1					

ClinVar clinical significance of the dbSNP variant

P1 0.0387 benign - -

- 0.0387 benign - -

P1 0.0387 benign - -

- 0.0387 benign - -

P1 0.0387 benign - -

P1 0.0387 benign - -

P1 0.0387 benign - -

- 0.0387 benign - -

- 0.0387 benign - -

P1 0.0377 likely_benign 27863249 38730892 -

OMIM- Online Mendelian Inheritance in Man

- <https://www.ncbi.nlm.nih.gov/omim> (create bookmarks!)
- Open access
- Listings of genes associated with human disease or syndrome
- Authored summaries of primary research articles

Search: USH2A Gene

- Lists syndromes associated with the gene
- Links, links and more links
- Info on **some** reported variants in “Allelic Variants”
- Note the edit history

Table of Contents for *608400

- Title
- Gene-Phenotype Relationships
- Text
- Cloning and Expression
- Gene Structure
- Mapping
- Gene Function
- Molecular Genetics
- Genotype/Phenotype Correlations
- Animal Model
- Allelic Variants
- Table View
- References
- Contributors
- Creation Date
- Edit History

MIMmatch (login)

***608400**
USH2A GENE; USH2A

Alternative titles; symbols
USHERIN
USH2; US2

*HGNC Approved Gene Symbol: **USH2A***

*Cytogenetic location: **1q41** Genomic coordinates*
*(**GRCh38**): **1:215,622,893-216,423,395** (from NCBI)*

Gene-Phenotype Relationships

Location	Phenotype	Phenotype MIM number	Inheritance (in progress)	Phenotype mapping key
1q41	Retinitis pigmentosa 39	613809		3
	Usher syndrome, type 2A	276901	AR	3

TEXT
Cloning and Expression
From the Usher syndrome type IIa (**276901**) critical region on chromosome 1 established by linkage studies, [Eudy et al. \(1998\)](#) identified the USH2A gene and

External Links

- Genome
- DNA
- Protein
- Gene Info
- Clinical Resource
- Variation
- Animal Models
- Cellular Pathways

Search: Noonan syndrome

- **Clinical synopsis**
 - List of features (by body system)
- Alternative names

Table of Contents for #163950

- Title
- Phenotype-Gene Relationships
- Text
- Description
- Clinical Features
- Other Features
- Inheritance
- Population Genetics
- Mapping
- Cytogenetics
- Diagnosis
- Clinical Management
- Molecular Genetics
- Nomenclature
- Animal Model
- History
- Phenotypic Series
- Clinical Synopsis
- See Also
- References
- Contributors
- Creation Date
- Edit History

MIMmatch (login)

#163950
NOONAN SYNDROME 1; NS1

Alternative titles; symbols
NOONAN SYNDROME
MALE TURNER SYNDROME
FEMALE PSEUDO-TURNER SYNDROME
TURNER PHENOTYPE WITH NORMAL KARYOTYPE

Other entities represented in this entry:
PTERYGIUM COLLI SYNDROME, INCLUDED

Phenotype-Gene Relationships

Location	Phenotype	Phenotype MIM number	Inheritance (in progress)	Phenotype mapping key	Gene/Locus	Gene/Locus MIM number
12q24.13	Noonan syndrome 1	163950	AD	3	PTPN11	176876

Clinical Synopsis**Phenotypic Series**

TEXT

A number sign (#) is used with this entry because Noonan syndrome-1 (NS1) is caused by heterozygous mutation in the PTPN11 gene (176876) on chromosome 12q24.

OMIM- Limitations/Potential Pitfalls

- Syndrome caused by multiple genes (like Noonan syndrome) has an entry for each gene
 - Difficult to get a snapshot of all the genes associated with a particular syndrome
- Should not replace a PubMed search
 - Edit history
 - Not complete
- Summaries are not for a lay audience

Slide borrowed from:

Bioinformatics Databases with Clinical Relevance

Natalie Carter, MS, LCGC

HUGEN 2070: Bioinformatic Resources for Geneticists

November 1, 2019

GeneReviews

- www.genereviews.org
- Open access
- Chapters written by experts and peer reviewed
 - Details for diagnosis
 - Genetic testing information
- Syndrome based
 - Favor toward syndromes with available (single gene) genetic testing

Slide borrowed from:

Bioinformatics Databases with Clinical Relevance

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HUGEN 2070: Bioinformatic Resources for Geneticists

November 1, 2019

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Molecular Genetic Testing Used in Noonan Syndrome (NS)

Gene ¹	Proportion of NS Attributed to Pathogenic Variants in This Gene	Proportion of Pathogenic Variants ² Detected by Test Method	
		Sequence analysis ³	Gene-targeted deletion/duplication analysis ⁴
<i>PTPN11</i>	50% ⁵	Nearly 100%	Rare duplication ⁶ , diagnosis of NS questioned ⁷
<i>SOS1</i>	10%-13% ⁸	100%	Unknown ⁹
<i>RAF1</i>	5% ¹⁰	Nearly 100%	One reported case with a duplication ¹¹ , diagnosis of NS questioned ⁷ One reported case of a deletion ¹²
<i>RIT1</i>	5% ¹⁰	100%	Unknown ⁹
<i>KRAS</i>	<5% ¹³	100%	Unknown ⁹
<i>NRAS</i>	8 individuals and 4 families ¹⁴	100%	Unknown ⁹
<i>BRAF</i>	<2% ¹⁵	100%	Unknown ⁹
<i>MAP2K1</i>	<2% ¹⁶	100%	Unknown ⁹
Others ¹⁷	NA		

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“Higher-throughput” annotation

How can a very large dataset be efficiently annotated?

That is, how can we combine *all* of the types of annotation we’ve discussed for a *large* number of variants (e.g., a VCF file from a sequencing project, with 5 million SNPs and 500,000 indels)?

One commonly-used tool is the command-line program ANNOVAR, which can annotate your variants with the data underlying essentially any track of the UCSC Genome Browser – or your own data.

ANNOVAR inputs and basic functions

ANNOVAR is an efficient software tool to utilize update-to-date information to functionally annotate genetic variants detected from diverse genomes (including human genome hg18, hg19, hg38, as well as mouse, worm, fly, yeast and many others). Given a list of variants with chromosome, start position, end position, reference nucleotide and observed nucleotides, ANNOVAR can perform:

- **Gene-based annotation:** identify whether SNPs or CNVs cause protein coding changes and the amino acids that are affected. Users can flexibly use RefSeq genes, UCSC genes, ENSEMBL genes, GENCODE genes, AceView genes, or many other gene definition systems.
- **Region-based annotation:** identify variants in specific genomic regions, for example, conserved regions among 44 species, predicted transcription factor binding sites, segmental duplication regions, GWAS hits, database of genomic variants, DNase I hypersensitivity sites, ENCODE H3K4Me1/H3K4Me3/H3K27Ac/CTCF sites, ChIP-Seq peaks, RNA-Seq peaks, or many other annotations on genomic intervals.
- **Filter-based annotation:** identify variants that are documented in specific databases, for example, whether a variant is reported in dbSNP, what is the allele frequency in the 1000 Genome Project, NHLBI-ESP 6500 exomes or Exome Aggregation Consortium (ExAC) or Genome Aggregation Database (gnomAD), calculate the SIFT/PolyPhen/LRT/MutationTaster/MutationAssessor/FATHMM/MetaSVM/MetaLR scores, find intergenic variants with GERP++ score<2 or CADD>10, or many other annotations on specific mutations.
- **Other functionalities:** Retrieve the nucleotide sequence in any user-specific genomic positions in batch, identify a candidate gene list for Mendelian diseases from exome data, and other utilities.

ANNOVAR

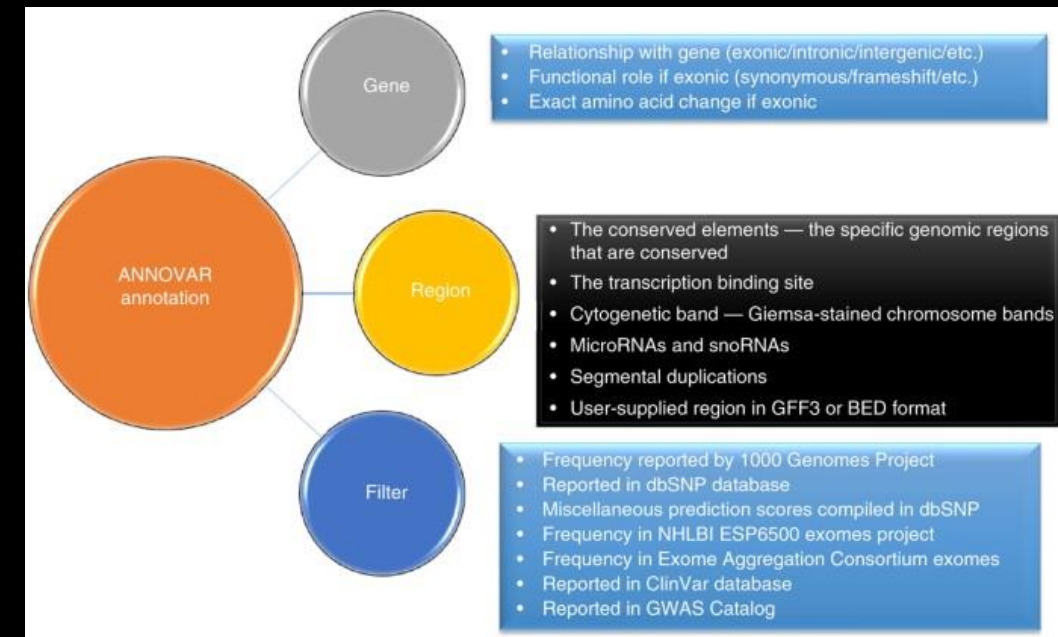
Website

<https://annovar.openbioinformatics.org/en/latest/>

Documentation

<https://academic.oup.com/nar/article/38/16/e164/1749458>

<https://www.nature.com/articles/nprot.2015.105>



To run ANNOVAR, you have to locally save copies of the annotation databases you want to use (ANNOVAR will do this for you).

Input: text-base file with one line per variant. Must include chromosome, start/end positions, and alleles.

Types of annotation:

- Gene-based (e.g., RegSeq, UCSC, ENSEMBL, GENCODE)
- Region-based (e.g., conserved regions, chip-seq data, transcription factor binding sites, etc.)
- Filter-based (i.e., matching variants to annotation based on exact position/alleles, and filtering on the annotation values)

ANNOVAR

ANNOVAR consists of several Perl scripts, some of which have multiple functions

- **annotate_variation**

- # Download database

- # Optional arguments include genome build

- # Database examples: gene/refGene; 1000genomes; snp136; Cytoband; tfbsConsSites

- > perl annotate_variation.pl --downdb [options] <table_name> <output_path>

- **convert2annovar**

- # Use this to convert VCF, Samtools, or other filetypes to the correct input format

- > perl convert2annovar.pl old_file.vcf --format vcf4 > new_file.vcf4

ANNOVAR

- **annotate_variation**

#Use this script again to do gene-based annotation

Output = 2 files, one with non-exonic variant function (upstream, UTR, etc.), another with exonic function (e.g., nonsynonymous)

> perl annotate_variation.pl new_file.vcf4 --buildver hg19 annotation_database_directory/

- **annotate_variation**

Use this script again to do filter-based annotation (match by exact location+alleles)

Output = 2 files, one with known variants, the other with unmatched variants

You can also filter out variants based on the values of the annotate data

> perl annotate_variation.pl new_file.vcf4 --filter --dbtype 1000g2012apir_EUR --buildver hg19 annotation_database_directory/

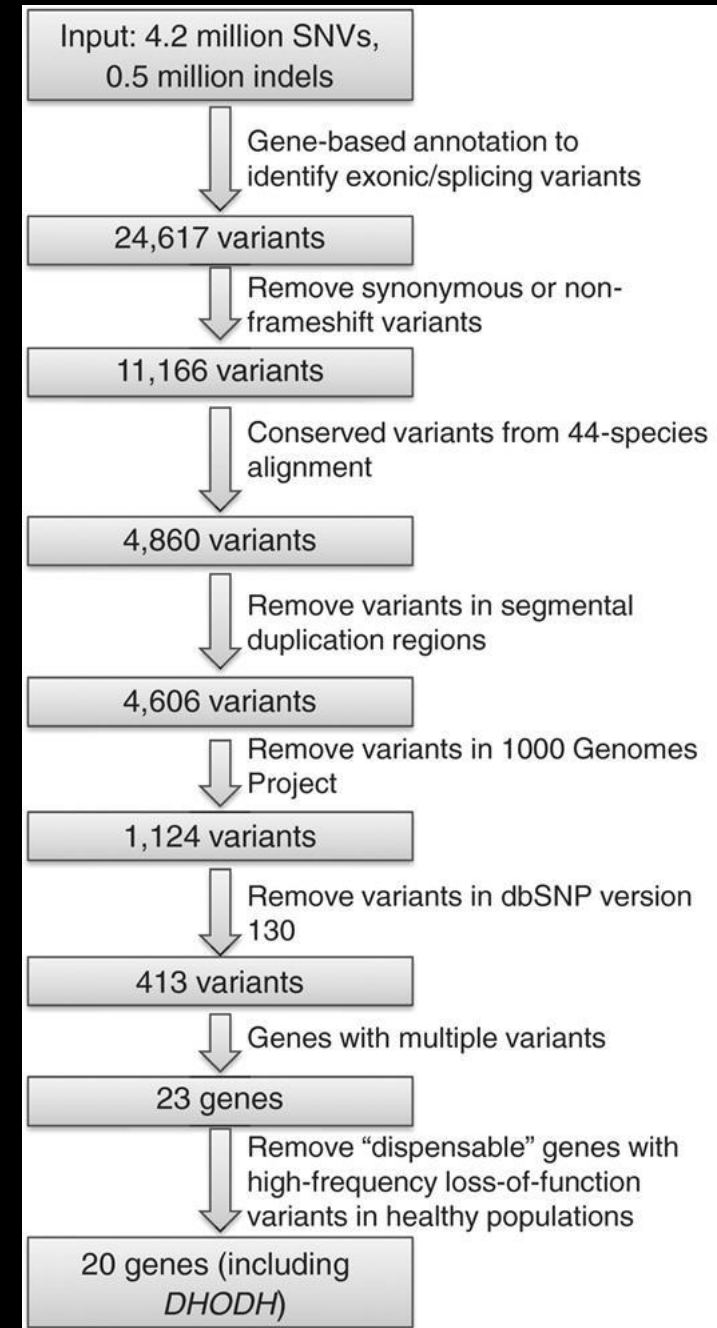
ANNOVAR

- **annotate_variation**
Use this script *again* to do region annotation
Output: annotated file
> perl annotate_variation.pl new_file.vcf4 -regionanno -dbtype <argument> -buildver hg19
[more options] annotation_database_directory/
- **summarize_annovar**
Integrates results from multiple databases and outputs them to an Excel file
- **variants_reduction**
Allows more advanced/customized filtering options across multiple values across multiple databases, with the goal of helping you prioritize/identify causal variants
- **retrieve_seq_from_fasta**
Retrieve base sequence for region
- **coding_change**
Retrieves protein sequence (for “reference” vs for “mutated” genome)

ANNOVAR vignette

Example from ANNOVAR paper

- Try to find a Mendelian disease gene in an individual's sequence data
- Approach: Take sequence data from “unaffected” people and add in known Mendelian disease variants to “simulate” the disease
- Use a simple annotation/filtering procedure to whittle down the list of genes and try to come up with the causal genes
- The example at the right uses 1 subject to get to 20 genes (Note that PolyPhen calls one of the causal mutations “benign”)
- In a different example, intersecting gene lists from 4 people
 - 1 subject → 159 genes
 - 2 subjects → 13 genes
 - 3 subjects → 6 genes
 - 4 subjects → 4 genes (including the causal gene)
 - SIFT + PolyPhen → 2 genes (including causal gene)



wANNOVAR

<https://wannovar.wglab.org>

 wANNOVAR

[Home](#) [Tutorial](#) [Example](#) [Related projects](#)

 WGLAB

wANNOVAR

ANNOVAR is a rapid, efficient tool to annotate functional consequences of genetic variation from high-throughput sequencing data. wANNOVAR provides easy and intuitive web-based access to the most popular functionalities of the ANNOVAR software

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By default, wANNOVAR performs "individual analysis" on the first sample in your VCF file to help find disease genes (you may need to split your multi-sample VCF file to individual files for annotation separately to find disease genes). If you only want to annotate all variant sites in a multi-sample VCF file, select "All Annotations" option below.

If you see an error message "cannot create submission directory for submission: no space left", it means our server's storage space is full. You can wait 1-2 days to submit again, or you can email Kai to resolve the issue quickly.

Basic Information

Email	Email
Sample Identifier	Sample Identifier
Input File	+ Input File
or Paste Variant Calls	paste your variant call here

Recent Updates

[5/9/2019] The wANNOVAR server is migrated to a new host. All submissions will be deleted **within 1 day (new rule in June 2019)** due to lack of storage space. Please download your results promptly.

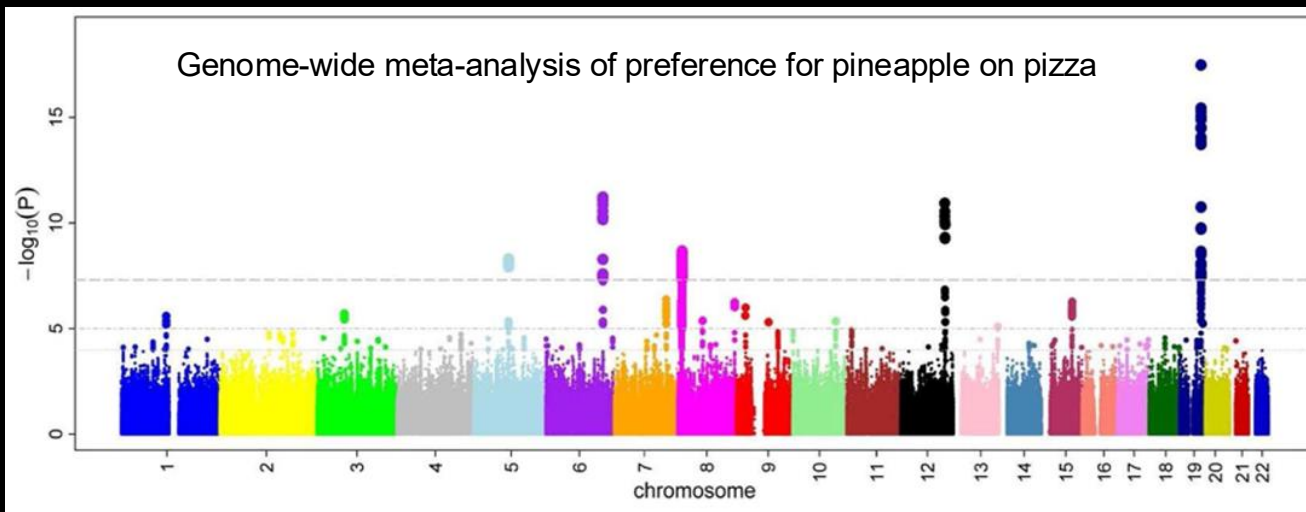
[10/19/2017] The detailed amino acid changes for indels are now included in the output (through - polish argument in table_annotvar). The server also handles duplicated entries (multiple identical variants) in the input file correctly.

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3556337/>

Some guidelines for interpreting evidence

What to do now with all this annotation?

You should just publish the plot and the table, right? (No.)



SNP	P	Annot1	Annot2	Annot3	...	Annot738
chr1:135367430	2.4E-01	GENE1	Missense	Pathog.	...	104.633
chr1:111988570	1.7E-02	GENE2	Intronic	Benign	...	57245.16
chr1:112070215	5.4E-01	GENE3	Synon.	Benign	...	12456.33
chr1:112091824	1.0E-01	GENE4	Synon.	Benign	...	47245.
chr2:102082746	9.4E-01	GENE5	Intronic	Benign	...	3445723.3
chr2:102086444	4.7E-01	GENE6	Intronic	Benign	...	35234.356
chr2:102118534	3.5E-01	GENE7	Missense	Pathog.	...	4234.65
chr2:102129814	4.9E-01	GENE8	5'UTR	Benign	...	347.125
...	...				...	
...	...				...	
chr22:14567264	9.2E-03	Gene25	Synon.	Benign	...	1347.36

Approaching the interpretation of evidence

It's likely that for any given association there won't be any single piece of annotation that reasonably leads you to think, "Aha! This association must be 'real' and this must be the causal SNP!"

Somehow you have to decide whether and how the layers of annotation "stack up" to interesting evidence about a variant.

Approaching the interpretation of evidence

An additional layer of complexity comes from the fact that the “calibration” of some of the tools may be unclear. For instance, if you come up with a ranking system similar to RegulomeDB’s method, how can you know if you’re really prioritizing based on an important metric? What if tons of SNPs are near histone marks and are in LD with eQTLs? Probably you should look for evidence that somehow ties in with what you already know or have already hypothesized about about your trait.

And on the other hand, you shouldn’t necessarily let a *lack* of interesting annotation convince you that a particular variant must be irrelevant. Why?

Approaching the interpretation of evidence

Since your study was probably associational, you probably can't infer causality. But your annotation may generate further testable hypotheses that could inform you about mechanisms/causality. To do that, you probably need to try to extend what you already know about your trait in a simple way based on your results.

One suggestion I have: read some genetic association study papers, maybe from journals with very different “impact factors”. Try to concisely summarize what evidence the authors present and the flow of their argument in just a few sentences. How convincing do you find the arguments supporting their conclusions/hypotheses?

Approaching the interpretation of evidence

Fine-mapping/prioritization tools

Some tools integrate association results with annotation tracks and prioritize the association results (e.g., by computing a posterior probability of a SNP being causal).

PAINTOR (<https://journals.plos.org/plosgenetics/article?id=10.1371/journal.pgen.1004722>) is one example.

Suppose you run your results through such a tool and obtain a prioritized list. What are some questions you might have about the results?

Tangent: gene names

The same gene can have a lot of different names.

If you're doing a literature search, you may also want to search for its aliases.

Aliases for SLC39A10 Gene	
Aliases for SLC39A10 Gene	
Solute Carrier Family 39 Member 10 ^{2 3 4 5}	ZIP-10 ^{3 4}
Solute Carrier Family 39 (Metal Ion Transporter), Member 10 ^{2 3}	DKFZp564L2123 ²
Solute Carrier Family 39 (Zinc Transporter), Member 10 ^{2 3}	SLC39A10 ⁵
Zrt- And Irt-Like Protein 10 ^{3 4}	FLJ90515 ²
Zinc Transporter ZIP10 ^{3 4}	LZT-Hs2 ³
KIAA1265 ^{2 4}	ZIP10 ⁴

Tangent: gene names

The most interesting result doesn't come up in the search for the official gene name.

PubMed.gov

slc39a10 oocyte

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RESULTS BY YEAR

TEXT AVAILABILITY

ARTICLE ATTRIBUTE

ARTICLE TYPE

1 Zinc supplementation alleviates endoplasmic reticulum stress during porcine **oocyte** in vitro maturation by upregulating zinc transporters.
Cite Ridlo MR, Kim GA, Taweetchaipaisankul A, Kim EH, Lee BC.
J Cell Physiol. 2021 Apr;236(4):2869-2880. doi: 10.1002/jcp.30052. Epub 2020 Sep 17.
PMID: 32944961

2 A candidate gene analysis and GWAS for genes associated with maternal nondisjunction of chromosome 21.
Cite Chernus JM, Allen EG, Zeng Z, Hoffman ER, Hassold TJ, Feingold E, Sherman SL.
PLoS Genet. 2019 Dec 12;15(12):e1008414. doi: 10.1371/journal.pgen.1008414. eCollection 2019 Dec.
PMID: 31830031 Free PMC article.

3 Cloning, function, and localization of human, canine, and Drosophila ZIP10 (SLC39A10), a Zn(2+) transporter.
Cite Landry GM, Furrow E, Holmes HL, Hirata T, Kato A, Williams P, Strohmaier K, Gallo CJR, Chang M, Pandey MK, Jiang H, Bansal A, Franz MC, Montalbetti N, Alexander MP, Cabrero P, Dow JAT, DeGrado TR, Romero MF.
Am J Physiol Renal Physiol. 2019 Feb 1;316(2):F263-F273. doi: 10.1152/ajprenal.00573.2017. Epub 2018 Dec 6.
PMID: 30520657 Free PMC article.

4 Oocyte-cumulus cell interactions regulate free intracellular zinc in mouse **oocytes**.
Cite Lisle RS, Anthony K, Randall MA, Diaz FJ.
Reproduction. 2013 Apr 15;145(4):381-90. doi: 10.1530/REP-12-0338. Print 2013 Apr.
PMID: 23404848

PubMed.gov

zip10 oocyte

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RESULTS BY YEAR

TEXT AVAILABILITY

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ARTICLE TYPE

1 Cloning, function, and localization of human, canine, and Drosophila **ZIP10** (SLC39A10), a Zn(2+) transporter.
Cite Landry GM, Furrow E, Holmes HL, Hirata T, Kato A, Williams P, Strohmaier K, Gallo CJR, Chang M, Pandey MK, Jiang H, Bansal A, Franz MC, Montalbetti N, Alexander MP, Cabrero P, Dow JAT, DeGrado TR, Romero MF.
Am J Physiol Renal Physiol. 2019 Feb 1;316(2):F263-F273. doi: 10.1152/ajprenal.00573.2017. Epub 2018 Dec 6.
PMID: 30520657 Free PMC article.

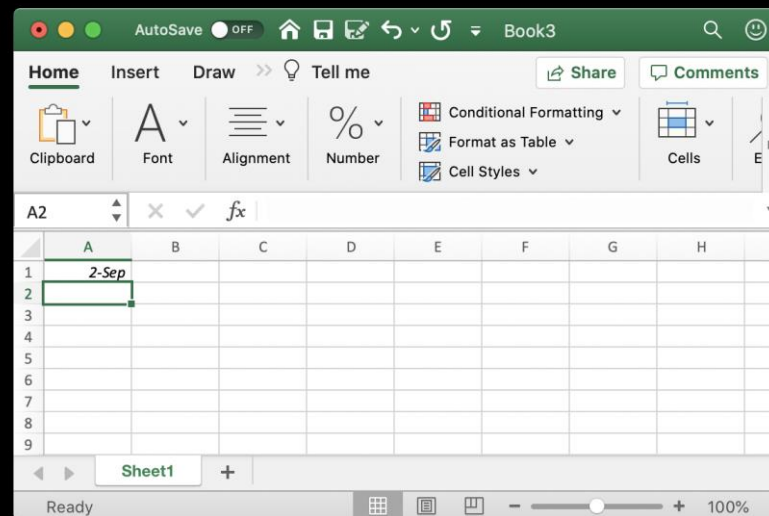
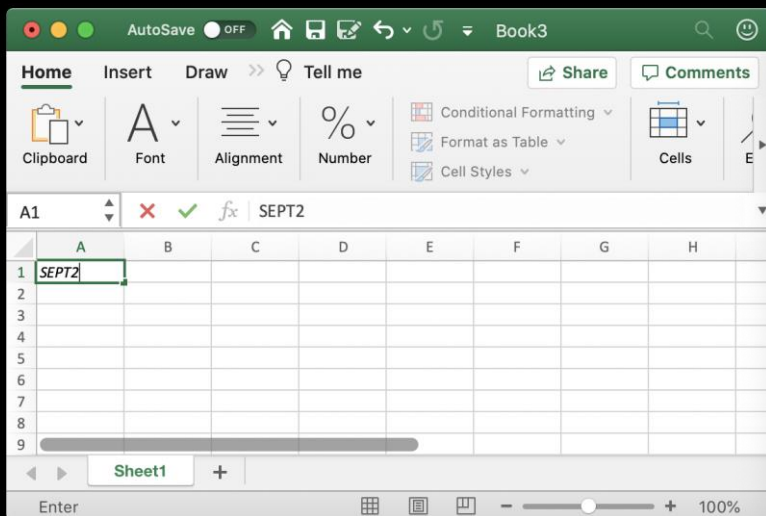
2 Maternally-derived zinc transporters ZIP6 and **ZIP10** drive the mammalian **oocyte-to-egg** transition.
Cite Kong BY, Duncan FE, Que EL, Kim AM, O'Halloran TV, Woodruff TK.
Mol Hum Reprod. 2014 Nov;20(11):1077-89. doi: 10.1093/molehr/gau066. Epub 2014 Aug 20.
PMID: 25143461 Free PMC article.

3 Cadmium-induced ovarian pathophysiology is mediated by change in gene expression pattern of zinc transporters in zebrafish (Danio rerio).
Cite Chouchene L, Banni M, Kerkeni A, Saïd K, Messaoudi I.
Chem Biol Interact. 2011 Sep 5;193(2):172-9. doi: 10.1016/j.cbi.2011.06.010. Epub 2011 Jul 3.
PMID: 21756885

4 Regulation of ZIP and ZnT zinc transporters in zebrafish gill: zinc repression of **ZIP10** transcription by an intronic MRE cluster.
Cite Zheng D, Feeney GP, Kille P, Hogstrand C.
Physiol Genomics. 2008 Jul 15;34(2):205-14. doi: 10.1152/physiolgenomics.90206.2008. Epub 2008 May 13.
PMID: 18477665 Free PMC article.

Tangent: gene names

Excel thinks this is a date



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Gene name errors are widespread in the scientific literature

[Mark Ziemann](#), [Yotam Eren](#) & [Assam El-Osta](#) [✉](#)

[Genome Biology](#) 17, Article number: 177 (2016) | [Cite this article](#)

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Abstract

The spreadsheet software Microsoft Excel, when used with default settings, is known to convert gene names to dates and floating-point numbers. A programmatic scan of leading genomics journals reveals that approximately one-fifth of papers with supplementary Excel gene lists contain erroneous gene name conversions.

Aliases for SEPTIN2 Gene

Aliases for SEPTIN2 Gene

Septin 2 ^{2 3 5}
Neural Precursor Cell Expressed, Developmentally Down-Regulated 5 ^{2 3}
Septin-2 ^{3 4}
KIAA0158 ^{2 4}
NEDD-5 ^{3 4}
Pnutl3 ^{2 3}

HNedd5 ^{2 3}
DIFF6 ^{3 4}
NEDD5 ^{3 4}
SEPT2 ^{3 4}
Neural Precursor Cell Expressed Developmentally Down-Regulated Protein 5 ⁴
Epididymis Secretory Sperm Binding Protein ³
SEPTIN2 ⁵

External Ids for SEPTIN2 Gene

HGNC: [7729](#) Entrez Gene: [4735](#) Ensembl: [ENSG00000168385](#) OMIM: [601506](#) UniProtKB: [Q15019](#)

Previous HGNC Symbols for SEPTIN2 Gene

DIFF6, NEDD5, **SEPT2**